Università degli studi di Milano
Dipartimento di Fisica

Ongoing notes

Cosmology & Astrophysics

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Abstract

These are never-ending notes where I teach myself stuff. Hopefully they will grow into something useful, possibly also for other people. We should cite something [ACRS22]

Statement of Non-Originality

These notes are far from being original. It is just my personal rearrangement of the subject as I continue to learn it from a variety of people (much smarter than me).

Joining the project

If you find these notes useful, if you find typos or if you would like to collaborate or suggest chages, do get in touch!

README: Collaborating, conventions and examples (in progress)

Conventions for Notes, Questions, TODOs, Problems etc

- Use `\answertag{...}`, `\notetag{...}`, `\questiontag{...}`, `\todotag{...}` for page margin notes (green, yellow, orange, red, respectively). As a rule of thumb
  - `\answertag{...}` for answer
  - `\notetag{...}` for notes or minor problems
  - `\notetag{...}` for questions or problems
  - `\notetag{...}` for things to be done or big problems
- Use `\answerinline{...}`, `\noteinline{...}`, `\questioninline{...}`, `\todoinline{...}` for inline notes (for equations or wherever you like).
- For brevity, use (cite) to denote missing citations and (*ref) to denote missing references to labels.

Examples

This is some text that needs some note. More text that raises some questions.
This sentence contains an inline note Check this value! and continues without breaking the line. And now some more text with some answers. Next we have an example in math mode

A margin note /minor problem

A question or problem

An answer.

$$E = mc^2. \quad \text{Explain this equation is often false.}$$

Conventions for Labelling

Labeling is in snake_case in the form `\label{TYPE:name_in_snake_case}` with some meaningful name.

- Sections, subsections etc are all labelled with `\label{sec:name_of_section`
- Equations are labelled with `\label{eq:name_of_equation}`
- Figures are labelled with `\label{fig:name_of_equation}`
- And so on . . .

a Todo or big problem

Something else that should be decided beforehand

I might very well have forgotten something. . .

Introduction

Cosmology is the study of the Universe as a whole, its origin, evolution, structure, and eventual fate...

Structure of the work.

The notes are organized as follows. . .

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Pending points

Random pending points

- cannibalism in DM models (in advanced section)
- lensing of CMB
- appendix of Julien paper on cannibalism: entropy done right, and $m - \mu \gg T$ done right
- paper by Buen Amad on cannibalism: rhs of boltzmann eq for momenta done right

Whinter & Dodelson

1. L1-6 finish boltzmann bkg equations
2. L7-8 finish visibility function
3. RHS of Boltzmann eqs for phtons (dodelson)
4. winther L15 form of $T_\mu \nu$ for photons, projecting doewn T_{ij} on scalar part (trace and traceless), showing the effect of anisotropic stress from neutrinos and photons
5. whinter L16 initial conditions from inlfation (last part)
6. whinter L17 Finish this
   - spherical harmonics (ℓ is the scale of θ variation, m is the scale of ϕ variation etc)
   - stochastic radnom field in real space are always independent in Foruier space reagrdless of their correlation functions in real space. Homogeneity demands correlation function in real space to depend only on $|\vec{x} - \vec{y}|$, and this enforces total omentum conserveation (cf. manybody)
   - correlation function of matter powerspectrum: form and bao feature at 150MPc–> bao oscillations in fourier space
7. whinter L18 (finish getting $a_{\ell m}$ coefficients, and getting the power spectrum C_ℓ)

Part I

Generalities, Observations & Instruments

1 Generalities, Observations & State of the Research

1 General Principles of Cosmological Data Analysis

1.1 Ergodic princple & Cosmic variance

Ergodic principle
The typical theoretical observable in Physics is a stochastic process

$$X : \Omega \times U \to F \mid \omega, u \mapsto X(\omega, u),$$

where $(\Omega, \mathbb{P})$ is the probability space of possible realizations, (U, μ_U) is the space of parameters endowed with some measure μ_U, and F is the phase space for possible values of the observable. The space U is typically the spacetime manifold, or a subset of it like a spatial or temporal slice, and the measure μ_U is typically the Lebesgue measure.

Theory predicts the law of the stochastic process i.e. the probability distribution induced on maps $U \to F$ by X, while experiments can only access a single realization of the process, i.e. a single map $U \to F$.

To compare theory and experiments, we must either repeat the experiment several times (i.e. draw several realizations of the process) or we must assume a single realization of the process is enough to explore the whole phase space, so that we can replace ensemble averages with averages over (a subset of) the parameter space U like spatial or temporal averages.

The ergodic principle indeed postulates that ensemble averages are independent of the process parameters u and are equal to averages over the parameter space,

$$\mathbb{E}_\Omega\big[X(\cdot, u_*)\big] = \langle X(\cdot, u_*)\rangle_\Omega \stackrel{!}{=} \int_U d\mu_U \; X(\omega_*, u), \quad \forall u_* \in U, \omega_* \in \Omega.$$

The parameter space is often just space or time depending on the context, and parameter averages are spatial or temporal averages. For example in Cosmology we only consider spatial averages, since laws of observables do vary over time and in any case temporal variations would not be observable over the lifetime of man kind. On the other hand in statistical mechanics we only consider temporal averages, since laws of microphysical processes do not change over time and we can observe temporal variations thereof. In any case, for the ergodic principle to hold, it is necessary the stochastic process be stationary, i.e. the law of $X(\cdot, u)$ does not depend on u, and sufficiently uncorrelated on large scales, i.e.

$$\lim_{|u-u'|\to\infty} \langle X(\cdot, u) X(\cdot, u')\rangle_\Omega = \langle X(\cdot, u)\rangle_\Omega \, \langle X(\cdot, u')\rangle_\Omega, \quad \forall u, u' \in U.$$

One can prove this assumption is always satisfied for stationary Gaussian processes. However nonlinearities of the theory will drive Gaussian initial conditions towards non-gaussian distributions which might fail the assumptions. For example in Cosmology gravity will eventually talk across the Universe and make far away regions correlated.

Cosmic variance
Cosmic variance is the statistical uncertainty inherent to the fact that we replace ensemble averages with parameter averages of a single realization, pretending that $X(\omega_*, u)$ and $X(\omega_*, u')$ are uncorrelated for sufficiently large $|u - u'|$ and invoking the law of large numbers. This approximation is more accurate the higher the number of almost uncorrelated samples $X(\omega_*, u)$ we can fit in the parameter space U, i.e. essentially the larger the parameter space U and the smaller the correlation length (be it spatial or temporal) of the process are.

Consider first a toy model. Suppose theory predicts a given observable O is uniformly distributed in the interval $[0, 1]$ and we can measure only 5 values of O say

$$O(\omega_*, u_1) = 0.2, \quad O(\omega_*, u_2) = 0.6, \quad O(\omega_*, u_3) = 0.3, \quad O(\omega_*, u_4) = 0.4, \quad O(\omega_*, u_5) = 0.5.$$

By the ergodic principle we would estimate the mean of O as $\langle O\rangle_\Omega \approx 0.4$ and the variance as $\sigma_O^2 \approx 0.025$, whereas theory predicts $\langle O\rangle_\Omega = 0.5$ and $\sigma_O^2 = 1/12 \approx 0.083$.

Now consider realistic cosmological observables such as CMB multipoles or galaxy clustering. Theory predicts the spherical harmonic coefficients $\{a_{\ell m}\}_{m=-\ell\dots\ell}$ of CMB temperature contrast be i.i.d. for fixed ℓ with a certain probability distribution. However we only measure $2\ell+1$ such values $a_{\ell m}^{\text{obs}} = a_{\ell m}(\omega_*)$ for the given realization ω^* of our universe, and then estimate CMB multipoles as

$$C_\ell^{\text{theory}} := \langle |a_{\ell m}^{\text{theory}}|^2 \rangle_\Omega \approx \frac{1}{2\ell+1} \sum_{m=-\ell}^{\ell} |a_{\ell m}^{\text{obs}}|^2 =: C_\ell^{\text{obs}}.$$

Similarly theory predicts a given stationary isotropic probability distribution for the matter overdensity field $\delta(\cdot, \vec{x})$ and define the correlation function as $\xi(r) := \langle \delta(\vec{x})\delta(\vec{x}+\vec{r})\rangle_\Omega$. However we only measure $\delta(\omega_*, \vec{x})$ for the given realization ω^* of our universe, and then estimate the correlation function as

$$\xi^{\text{theory}}(r) := \langle \delta(\cdot, \vec{x})\delta(\cdot, \vec{x}+\vec{r})\rangle_\Omega \approx \frac{1}{V}\int_V d^3x\ \delta(\omega_*, \vec{x})\delta(\omega_*, \vec{x}+\vec{r}) =: \xi^{\text{obs}}(r).$$

In both cases, the variance of the estimator is inversely proportional to the number of almost uncorrelated samples we can fit in the parameter space

$$\text{Var}(C_\ell^{\text{obs}}) \propto 1/(2\ell+1)\,, \qquad \text{Var}(\xi^{\text{obs}}(r)) \propto 1/V.$$

The question is then "how many azimuthal modes m can you fit in the sky for a given ℓ?" or "how many meters can you fit in the universe for a given correlation length r?".

2 Observational tensions

Despite its many predictions and successes, Λ CDM cosmology still exhibits tensions across different datasets; the most notable is the so-called H_0 tension [13, 14]. It appears when comparing local measurements of the expansion rate of the universe, such as those obtained by the SH0ES collaboration [15], with the value calculated by CMB experiments [5]. Another tension, albeit slightly milder, is the discrepancy that appears [16, 17] when measuring the clustering of small scale structure, often quantified with the S_8 parameter: in this case, weak lensing experiments such as DES [11, 18, 19], KiDS [12, 20-24], and CFHTLens [10] all find lower values of S_8 than those calculated by CMB experiments assuming Λ CDM cosmology. A recent analysis of [12] has obtained $S_8 = 0.762^{+0.025}_{-0.024}$, which is in 2.3σ tension with the Planck 2018 [5] value of $S_8 = 0.825 \pm 0.011$. We will adopt this measurement as our baseline S_8 data. We note, however, that by using slightly different assumptions and data, reference [22] gets a stronger (3.2σ) tension; while [24] also gets a $\sim 3\sigma$ tension when combining KiDS1000 weak lensing data with BOSS galaxy clustering data. These tensions have motivated the exploration of dark matter models beyond the standard CDM paradigm.

2.1 The H_0 tensions

3 Past, current & upcoming probes

2 Observations of the CMB

1 Experiments

Part II

Fiducial Cosmology

3 Isotropic and Homogeneous Universe

1 Cosmological Principle & FLRW Metric

1.0.1 A simple derivation of constant curvature spaces

See baumann

The **Cosmological Principle** states that the Universe is homogeneous and isotropic on large scales. The properties of the Universe are thus same everywhere and in all directions on sufficiently large scales (greater than about 100 Mpc). The CMB is the main experimental evidence for isotropy and LSS surveys support homogeneity.

Isotropy and homogeneity imply the sectional curvature of each 2d tangent plane at each point is the same. The spatial sections are thus constant-curvature 3d spaces, classified by the curvature K. The most general metric consistent with the Cosmological Principle is the **Friedmann-Lemaître-Robertson-Walker (FLRW) metric**

$$ds^2 = -dt^2 + a^2(t)\left[\frac{dr^2}{1-Kr^2} + r^2\underbrace{(d\theta^2 + \sin^2\theta d\phi^2)}_{d\Omega^2}\right] = -dt^2 + a^2(t)\left[d\chi^2 + S_K^2(\chi)d\Omega^2\right],$$

The relation between the comoving distance χ and the metric distance r is given by

$$r(\chi) = \begin{cases} \frac{1}{\sqrt{K}}\sin(\sqrt{K}\chi) & K > 0 \\ \chi & K = 0 \\ \frac{1}{\sqrt{|K|}}\sinh(\sqrt{|K|}\chi) & K < 0 \end{cases},$$

The Christoffel symbols for the FLRW metric are

$$\Gamma^0_{ij} = a\dot{a}\gamma_{ij}, \quad \Gamma^i_{0j} = \frac{\dot{a}}{a}\delta^i_j, \quad \Gamma^i_{jk} = \tilde{\Gamma}^i_{jk},$$

where $\tilde{\Gamma}^i_{jk}$ are the Christoffel symbols of the spatial metric γ_{ij}.

The Friedmann equations governing the dynamics of the scale factor $a(t)$ are

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{K}{a^2} + \frac{\Lambda}{3},$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p) + \frac{\Lambda}{3}.$$

The Hublle constant today is

$$H_0 = 100h\ \mathrm{km\,s^{-1}\,Mpc^{-1}} \approx\ h\,3.3\times 10^{-3}\,\mathrm{Mpc^{-1}}$$

The Friedmann equation are written in terms of the density parameters as

$$H^2 = H_0^2\left[\sum_x \Omega_x a^{-3(1+w_x)} + \Omega_K a^{-2} + \Omega_\Lambda\right],$$

where

$$\Omega_r = \frac{8\pi G\rho_{r0}}{3H_0^2}, \quad \Omega_m = \frac{8\pi G\rho_{m0}}{3H_0^2}, \quad \Omega_\Lambda = \frac{\Lambda}{3H_0^2}, \quad \Omega_K = 1 - \sum_x \Omega_x - \Omega_\Lambda.$$

1.1 Distances in Cosmology

1.2 Back of the envelope calculations

The age of the Universe Finsh

Constraining the cosmological parameters Ω_x we can get an estimate for the age of the universe from Friedman equation

$$\frac{(da/dt)}{a} = H_0\Big[\sum_x \Omega_x a^{-3(1+w_x)}\Big]^{1/2} \quad \Rightarrow \quad T_{\text{uni}} = \int dt = \int_0^1 \frac{da}{aH_0\Big[\sum_x \Omega_x a^{-3(1+w_x)}\Big]^{1/2}}.$$

Inserting approximate values we ger $T_{\text{uni}} \approx 13.8$ Gyr.

Finsh **Size of the ovserbale universe: particle horizon vs Hubble radius**

Similarly we can estimate the size of the observable universe as the comoving distance traveled by a photon since the Big Bang, i.e. the particle horizon

$$\dots$$

For polynomial expansion $a(t) \sim t^\beta$ we find this is similar to the Hubble radius $d_H = c/H(t)$ up to a factor $\beta/(1-\beta)$.

1.3 Fine tuning problems

The cosmological constant problem Polish

The effect of a cosmological constant in the EH action $\int\sqrt{-g}(R - 2\Lambda^{\text{bare}})$ can be reabsorbed as a vacuum energy contribution to the stress-energy tensor

$$G_{\mu\nu} + \Lambda^{\text{bare}} g_{\mu\nu} = 8\pi G T_{\mu\nu} \quad \Rightarrow \quad G_{\mu\nu} = 8\pi G\left(T_{\mu\nu} + T^\Lambda_{\mu\nu}\right), \quad T^\Lambda_{\mu\nu} = -\frac{\Lambda^{\text{bare}}}{8\pi G} g_{\mu\nu}\,.$$

Quantum vacuum energy also contributes to the stress tensor as a cosmological constant Λ_{vac} i.e. $\rho_{\text{vac}} = -\frac{\Lambda_{\text{vac}}}{8\pi G} = -p_{\text{vac}}$, the total effective cosmological constant is then

$$\Lambda_{\text{eff}} = \Lambda_{\text{bare}} + \Lambda_{\text{vac}}\,.$$

The theoretical prediction for vacuum energy density from QFT is $\rho_{\text{vac}} \sim M_{\text{Pl}}^4 \sim 10^{76}\,\text{GeV}^4$, while the experimentally observed value from cosmology is $\rho_\Lambda = \Omega_\Lambda \rho_{\text{crit0}} \approx (0.7)\frac{3H_0^2}{8\pi G} \sim 10^{-47}\,\text{GeV}^4$. This discrepancy of about 123 orders of magnitude is known as the cosmological constant problem, representing a significant fine-tuning issue since the *bare* vacuum energy entering Einstein equations must cancel the *quantum* vacuum energy to an extraordinary degree of precision to yield the observed value of Λ_{eff}

$$\frac{\Lambda_{\text{bare}}}{\Lambda_{\text{vac}}} = -1 + \frac{\Lambda_{\text{eff}}}{\Lambda_{\text{vac}}} \simeq -1 + 10^{123}.$$

The equation of state of vacuum energy is also reason of debate. The interpretation as vacuum energy implies $w = -1$, but dynamical DE models with $w \neq -1$ are also possible. The current experimental value from Planck 2018 is $w = -1.03 \pm 0.03$.

The flatness problem

The horizon problem

2 Homogeneous Boltzmann Equation & Thermal History

Finish

In Newtonian mechanics the Boltzmann equation reads

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \frac{\vec{p}}{m}\cdot\partial_{\vec{x}} f + \vec{F}\cdot\partial_{\vec{p}} f = C[f]\,.$$

Making this covariant we have

$$\frac{df}{d\lambda} = \frac{dx^\mu}{d\lambda}\frac{\partial f}{\partial x^\mu} + \frac{dP^\mu}{d\lambda}\frac{\partial f}{\partial P^\mu} = C[f]\,,$$

where λ is an affine parameter along the particle's worldline with energy dimension $[\lambda] = [E]^{-2}$ *defined* to enforce the geodesic equation holds

$$P^\mu = \frac{dx^\mu}{d\lambda}, \quad \frac{dP^\mu}{d\lambda} + \Gamma^\mu_{\alpha\beta}P^\alpha P^\beta = 0\,.$$

2.0.1 Overview of interactions & their relevance

$$e^+ + e^- \leftrightarrow \gamma + \gamma \quad \text{pair production/annihilation}$$

This process is relevant at temperatures $T \gtrsim m_e \sim 0.5$ MeV, and thus for the thermal history of the universe before BBN. After electron-positron annihilation there is not enough energy to produce e^+e^- pairs, blocking the $\leftarrow$ direction, and soonafter there won't be enough positrons for the $\rightarrow$ direction. This interaction would indeed change number density and momentum bwteen the two components of the photon-baryon fluid but is blocked at later times.

$$\nu + \bar{\nu} \leftrightarrow e^+ + e^- \quad \text{neutrino decoupling}$$

This process is relevant at temperatures $T \gtrsim 1$ MeV, and thus for the thermal history of the universe before BBN, but negligible at later times.

$$e^- + p \leftrightarrow e^- + p(+\gamma) \quad \text{Coulomb scattering (+Bremsstrahlung)}$$

This process is relevant at all times, but especially at late times when it keeps electrons and protons tightly coupled, forming a single photon-baryon fluid.

$$n \rightarrow p + e^- + \bar{\nu}_e \quad \text{neutron decay}$$

Free neutrons are unstable and decay with a lifetime of about 15 minutes. This process would thus have plenty of time to occur from the end of inflation down to recombination ~ 380.000 years after the Big Bang. However, shortly aftern BBN all neutrons have either decayed, or are bound in nuclei and are thus stable, making this process negligible at later times.

3 Big Bang Nucleosynthesis

Finish

4 Recombination

Finish

The main process affecting recombination

$$\begin{aligned}
&e^- + p \leftrightarrow H + \gamma \quad \text{ionization/recombination} \\
&e^- + \gamma \leftrightarrow e^- + \gamma \quad \text{electron Compton-Thomson scattering} \\
&p + \gamma \leftrightarrow p + \gamma \quad \text{proton Compton-Thomson scattering} \\
&e^- + p \leftrightarrow e^- + p(+\gamma) \quad \text{Coulomb scattering (+Bremsstrahlung)}
\end{aligned} \tag{3.1}$$

Note that at temperatures relevant for recombination both proton and electron are highly nonrelativistic, the Compton cross-section $\sigma_X \propto m_X^{-2}$ for proton-photon scattering is thus suppressed by a factor $(m_e/m_p)^2 \sim 10^{-6}$ wrt electron-photon scattering and thus nonnegligible. It is the electron-photon scattering that keeps photons and electrons tightly coupled before recombination. On the other hand, Coulomb scattering is highly effeicient to keep electrons and protons tightly coupled, forming a single photon-baryon fluid.

Beware there is no vanilla "recombination" process, and other should be considered. In fact the naive recombination directly into the ground state is highly inefficient since the high-energy photon emitted would immediately ionize another hydrogen atom. One solution is to emit two photons in the recombination process

$e^- + p \to H + \gamma + \gamma$, but this is a higher order process, and thus suppressed with much smaller rate. Alternatively, hydrogen could recombine into excited states H^* and then emit another Lyman-α photon to reach the ground state. In formulas

$$\begin{aligned} &e^- + p \to H + \gamma + \gamma (\text{two photon emission}) \\ &e^- + p \Leftrightarrow H^* + \gamma \leftrightarrow H + \gamma + \gamma \\ &H^* \leftrightarrow H + \gamma \quad (\text{Lyman-}\alpha \text{ transition}) \,, \end{aligned}$$

Moreover we should consider processes involving the nonnegligible fraction of helium, also including intermediate steps with excited states,

$$\begin{aligned} &e^- + He^{++} \leftrightarrow He^+ + \gamma \\ &e^- + He^+ \leftrightarrow He + \gamma \,. \end{aligned}$$

The famous Peebles equations [**?**] take into account all these processes to accurately describe recombination, but ultimately give the same qualitative picture

....

The background level Boltzmann equation for the $e^- + p \leftrightarrow H + \gamma$ process reads

$$\frac{dn_e}{dt} + 3Hn_e = -\langle \sigma v \rangle \left(n_e n_p - n_H n_\gamma \frac{n_e^{\text{eq}} n_p^{\text{eq}}}{n_H^{\text{eq}} n_\gamma^{\text{eq}}} \right) \,, \tag{3.2}$$

where n_e, n_p and n_H are the number densities of free electrons, protons and neutral hydrogen atoms respectively, and $\langle \sigma v \rangle$ is the thermally averaged recombination cross-section times velocity.

In the Saha approximation we assume that the reaction (3.1) is strong enough $\Gamma \gtrsim H$ to maintain equilibrium, so that the rhs of (3.2) vanishes and we have

$$\frac{n_e n_p}{n_H n_\gamma} \simeq \frac{n_e^{\text{eq}} n_p^{\text{eq}}}{n_H^{\text{eq}} n_\gamma^{\text{eq}}} . \tag{3.3}$$

To proceed further we assume

$$\begin{aligned} &\text{charge neutrality} \quad n_e = n_p \,, \\ &\text{baryon number conservation} \quad n_b = n_p + n_H \,, \\ &\text{photons follow the equilibrium distribution} \quad n_\gamma = n_\gamma^{\text{eq}} \,. \end{aligned}$$

The latter assumption is justified since photons are still tightly coupled to electrons via Compton scattering and the ratio of phtons to baryyons is very huge $\eta_b = \frac{n_\gamma}{n_b} \sim 10^9$, so that the photon distribution is not significantly affected by the much less numerous baryons. At recombination temperatures $T \sim 1eV$ everything is nonrelativistic, so that we can use nonrelativistic equilibrium number densities with Boltzmann sppression factor. Finally we use

$$\begin{aligned} &\text{photon number is not conserved} \quad \mu_\gamma = 0 \\ &\text{chemical equilibrium} \quad \mu_H = \mu_H + \mu_\gamma = \mu_p + \mu_e \,, \\ &\text{binding energy} \quad \epsilon_0 := m_p + m_e - m_H = 13.6eV \,. \end{aligned}$$

Defining the ionization fraction $X_e = n_e/n_b$, from (3.3) we then obtain the Saha equation for the ionization fraction

$$\frac{1 - X_e}{X_e^2} = \frac{4\sqrt{2}\zeta(3)}{\sqrt{\pi}} \left(\frac{m_e T}{2\pi} \right)^{3/2} \frac{e^{-\epsilon_0/T}}{n_b} \,. \tag{3.4}$$

We can understand qualitatively recombination from (3.4). At high temperatures $T \gg \epsilon_0$ the exponential suppression is negligible and the rhs is very small due to the huge photon-to-baryon ratio, so that $X_e \simeq 1$ and the universe is fully ionized. As the temperature drops below the hydrogen binding energy $T \lesssim \epsilon_0$ the exponential suppression becomes important and the rhs grows large, so that $X_e \ll 1$ and the universe recombines. However, due to the huge photon-to-baryon ratio, recombination is delayed down to temperatures $T_{\text{rec}} \sim 0.3eV \sim \frac{1}{40}\epsilon_0$ since the high-energy tail of the photon distribution is still able to ionize hydrogen atoms.

When the temperature drops further, the reaction rate Γ becomes smaller than the Hubble expansion rate H and the Saha approximation breaks down. To accurately describe recombination in this regime we need to solve the full Boltzmann equation (3.2), taking into account all the relevant processes. With the same manipulations and assumptions as above, we can rewrite the Boltzmann equation in terms of the ionization fraction

$$\frac{dX_e}{dt} = -\langle \sigma v \rangle n_b \left(X_e^2 - (1 - X_e) \frac{n_e^{\text{eq}} n_p^{\text{eq}}}{n_H^{\text{eq}} n_\gamma^{\text{eq}}} \right) \,.$$

4.1 Optical depth & visibility function

Finish

The optical depth τ is defined as the integral of the scattering rate along the line of sight

$$\tau(z) = \int_0^z dz' \frac{dt}{dz'} n_e(z') \sigma_T c \,,$$

where σ_T is the Thomson cross-section. The visibility function $g(z)$ is defined as the probability density that a photon last scattered at redshift z, and is given by

$$g(z) = e^{-\tau(z)} \frac{d\tau}{dz}.$$

The visibility function peaks at the redshift of recombination $z_{\text{rec}} \sim 1100$, and its width Δz gives the thickness of the last scattering surface, which is about $\Delta z \sim 100$. The visibility function also includes the effect of reionization at later times, which creates a second peak at $z_{\text{reion}} \sim 10$ with much smaller amplitude. The normalization of the visibility function is such that $\int_0^\infty g(z) dz = 1$, so that it can be interpreted as a probability density function for the last scattering redshift of CMB photons.

4 Perturbation theory in Cosmology

Metric perturbations remain small

Treating metric perturbations to linear order is justified since they remain very small, even when energy-density perturbations become extremely non-linear. For example, compare the (dimensionless) gravitational potential[1] $\phi \sim \frac{GM}{R}$ and the density contrast $\delta\rho/\bar{\rho}$, relative to the cosmic average matter density today, for various astrophysical objects:

- Earth: $\phi \sim 10^{-9}, \quad \frac{\delta\rho}{\rho} \sim 10^{30}$
- Sun: $\phi \sim 10^{-6}, \quad \frac{\delta\rho}{\rho} \sim 10^{28}$
- Milky Way: $\phi \sim 10^{-6}, \quad \frac{\delta\rho}{\rho} \sim 10^{6}$
- Galaxy cluster: $\phi \sim 10^{-5}, \quad \frac{\delta\rho}{\rho} \sim 10^{3}$
- Black hole at Schwarzschild radius $R_S = 2GM$: $\phi \sim \frac{1}{2}, \quad \frac{\delta\rho}{\rho} \sim \infty$.

1 Gauge fixing

We will mainly follow the standard notation from [MB95]. The metric has signature $-+++$. The conformal time is denoted by τ and differentiation with respect to it is denoted by $\dot{f}$. The proper time (of a comoving observer) is denoted by t, is related to conformal time by $dt = a\, d\tau$ and differentiation with respect to it is denoted by f'. We use natural units $c = 1$, $\hbar = 1$ and $\kappa_B = 1$.

The perturbation to a homogeneous isotropic FLRW metric is written in full generality as

$$-\mathrm{d}s^2 := g_{\mu\nu}dx^\mu dx^\nu = a^2(\tau)\left[-(1+2A)\,\mathrm{d}\tau^2 + 2B_i\,\mathrm{d}x^i\,\mathrm{d}\tau + ((1+2C)\delta_{ij} + 2E_{ij})\,\mathrm{d}x^i\,\mathrm{d}x^j\right], \tag{4.1}$$

with SVT decomposition

$$\underbrace{A,\, C}_{\text{scalar}}, \quad B_i = \underbrace{\partial_i\tilde{B}}_{\text{scalar}} + \underbrace{\hat{B}_i}_{\text{vector}}, \quad E_{ij} = \underbrace{\partial_{\langle i}\partial_{j\rangle}\tilde{E}}_{\text{scalar}} + \underbrace{\partial_{(i}\hat{E}_{j)}}_{\text{vector}} + \underbrace{\bar{E}_{ij}}_{\text{tensor}},$$

where we use the shorthand

$$\partial_{\langle i}\partial_{j\rangle}\tilde{E} \equiv \left(\partial_i\partial_j - \frac{1}{3}\delta_{ij}\nabla^2\right)\tilde{E}, \qquad \partial_{(i}\hat{E}_{j)} \equiv \frac{1}{2}\left(\partial_i\hat{E}_j + \partial_j\hat{E}_i\right).$$

Newtonian Gauge

We consider the *conformal newtonian* gauge. In this gauge, vector and tensor modes are excluded from the beginning $\hat{B} = \hat{E} = \bar{E} = 0$, and the residual gauge freedom is used to kill the non-diagonal scalar terms $\tilde{B} = \tilde{E} = 0$. The metric is conventionally[2] written by calling $A = \psi$, $C = -\phi$

$$-\mathrm{d}s^2 := g_{\mu\nu}dx^\mu dx^\nu = a^2(\tau)\left[-(1+2\psi)\,d\tau^2 + (1-2\phi)\,\delta_{ij}\,dx^i dx^j\right]. \tag{4.2}$$

This gauge has a simple physical interpretation: in the Newtonian limit ψ plays the role of the gravitational potential, while ϕ is interpreted as a local perturbation to the scale factor. Upon ignoring anisotropic stress, Einstein equations will imply that $\psi \simeq \phi$.

[1] Identified with the Newtonian-gauge potential Φ

[2] Beware of different conventions in the literature, swapping $\phi \leftrightarrow \psi$ or their signs. Here we adopt the convention of Baumann [Bau22], `CLASS` [Les11] and [MB95] so that $\psi \approx \phi$ in the absence of anisotropic stress. Dodelson [DS25] uses $\phi \mapsto -\phi$.

The Christoffel symbols in this gauge (to linear order)

$$\begin{aligned} \Gamma^0_{00} &= \mathcal{H} + \dot{\psi}, & \Gamma^0_{ij} &= \mathcal{H}\delta_{ij} - \left[\dot{\phi} + 2\mathcal{H}(\phi+\psi)\right]\delta_{ij}, \\ \Gamma^0_{0i} &= \partial_i\psi, & \Gamma^i_{j0} &= \mathcal{H}\delta_{ij} - \dot{\phi}\delta_{ij}, \\ \Gamma^i_{00} &= \partial_i\psi, & \Gamma^i_{jk} &= -\partial_j\phi\,\delta_{ki} - \partial_k\phi\,\delta_{ji} + \partial_i\phi\,\delta_{jk}. \end{aligned} \tag{4.3}$$

2 Relativistic kinematics and the Boltzmann equation

We set the notation for relativistic kinematics and the Boltzmann equations. The covariant 4-momentum is $P^\mu = \frac{dx^\mu}{d\lambda}$ where $d\lambda$ is an affine parameter with $[\lambda] = E^{-2}$ *defined* so that the geodesic equation $\frac{dP^\mu}{d\lambda} = \Gamma^\mu_{\alpha\beta}P^\alpha P^\beta$ holds[3]. We have the invariant volumes in position and/or momentum space

$$\sqrt{-g}\,d^4x^\mu, \quad \sqrt{-g}\,d^4P^\mu, \quad \frac{d^4P_\mu}{\sqrt{-g}}, \quad d^4\,x^\mu d^4P_\mu.$$

Enforcing the on-shell condition, we have the invariant volume in momentum space

$$\frac{d^4P_\mu}{\sqrt{-g}}\,\delta_+(P^\alpha P_\alpha + m^2) = \frac{d^3P_i}{\sqrt{-g}}\frac{1}{2P^0},$$

and in turn the invariant volume in phase space

$$\sqrt{-g}\,d^4x^\mu\frac{d^3P_i}{\sqrt{-g}}\frac{1}{2P^0} = d^4\,x^\mu\frac{d^3P_i}{2P^0}.$$

Define the physical 3-momentum $\vec{p}$ measured by a comoving observer, by enforcing $p^2 := g_{ij}P^iP^j$ with direction $\hat{p}^i \propto P^i$ normalized such that $\hat{p}^i\hat{p}^j\delta_{ij} = 1$. We have $g_{\mu\nu}P^\mu P^\nu = -m^2$ and enforce the relativistic energy relation $E = \sqrt{p^2+m^2}$, so that

$$E^2 = -g_{00}(P^0)^2 - 2g_{0i}P^0P^i, \quad P^0_\pm = \frac{g_{0i}P^i \pm \sqrt{(-g_{00})E^2 + (g_{0i}P^i)^2}}{(-g_{00})}, \tag{4.4}$$

where $P^0 = P^0_+$ always denotes the positive sign solution. Solving to linear order in the metric perturbations (4.1) we relate covariant and physical quantities

$$P^i = \frac{p}{a}\,(1 - C - E_{mn}\,\hat{p}^m\hat{p}^n)\,\hat{p}^i, \quad P^0_\pm = \pm\frac{E}{a}(1-A) + \frac{p}{a}B_i\,\hat{p}^i.$$

Finally, for any metric parametrized as in (4.1), using (4.4) we recover the usual invariant momentum volume element

$$\frac{d^3P_i}{\sqrt{-g}}\frac{1}{2P^0} = \frac{d^3\vec{p}}{2E}.$$

Geodesic equation in Newtonian gauge

In the newtonian gauge (4.2) we have $A = \psi$, $B = E = 0$, $C = -\phi$ and thus

$$P^i = \frac{p}{a}(1+\phi)\hat{p}, \quad P^0 = \frac{E}{a}(1-\psi), \quad P_0 = -a(1+\psi)E, \quad P_i = a(1-\phi)p\hat{p}_i. \tag{4.5}$$

The metric potentials then obey (to first order in metric perturbations)

$$\frac{d\psi}{d\lambda} = \frac{E}{a}\dot{\psi} + \frac{p}{a}\hat{p}^j\partial_j\psi, \qquad \frac{d\phi}{d\lambda} = \frac{E}{a}\dot{\phi} + \frac{p}{a}\hat{p}^j\partial_j\phi.$$

[3] For massive particles it is simply $d\lambda = \frac{ds}{m}$.

Using the Christoffel symbols (4.3) we have the geodesic equations:

$$\begin{aligned}\frac{dP^0}{d\lambda} &= -\frac{E^2}{a^2}\left(\mathcal{H}(1-2\psi)+\dot{\psi}\right) - 2\frac{Ep}{a^2}\hat{p}^j\partial_j\psi - \frac{p^2}{a^2}\left(\mathcal{H}(1-2\psi)+\dot{\phi}\right),\\ \frac{dP^k}{d\lambda} &= -\frac{E^2}{a^2}\partial_k\psi - \frac{Ep}{a^2}\hat{p}^k\,2\left(\mathcal{H}(1-\psi+\phi)-\dot{\phi}\right) - \frac{p^2}{a^2}\left(\partial_k\phi - 2\hat{p}^k\hat{p}^j\partial_j\phi\right).\end{aligned} \tag{4.6}$$

Using the above relations, we obtain corresponding evolution equations for the physical quantities, already expressed in conformal time τ Check

$$\begin{aligned}\frac{dE}{d\tau} &= -\frac{p^2}{E}(\mathcal{H}-\dot{\phi}) - p\hat{p}^j\partial_j\psi,\\ \frac{dp}{d\tau} &= -p(\mathcal{H}-\dot{\phi}) - E\hat{p}^j\partial_j\psi,\\ \frac{d\hat{p}^i}{d\tau} &= \left(\hat{p}^i\hat{p}^j-\delta^{ij}\right)\left(\frac{E}{p}\partial_j\psi + \frac{p}{E}\partial_j\phi\right).\end{aligned} \tag{4.7}$$

2.1 Boltzmann equation

The phase space distribution $f_s(\tau,\vec{x},P^\mu)$ describes the probability of finding a particle, in a big collection of a given species s, with (on-shell) covariant momentum P^μ at spacetime location $(\tau,\bar{x})$. The relativistic Boltzmann equation (BE) describes the evolution of this distribution

$$\frac{df}{d\lambda} = \frac{\partial f}{\partial\tau}P^0 + \frac{\partial f}{\partial x^i}P^i + \frac{\partial f}{\partial P^\mu}\frac{dP^\mu}{d\lambda} = C[f], \tag{4.8}$$

where λ is the affine parameter for the geodesic equation and the collision term $C[f]$ accounts for the interactions of the species s.

For a general interaction between species a_i, b_j of the schematic form

$$a_1 + a_2 + \cdots + a_m \leftrightarrow b_1 + b_2 + \cdots + b_n,$$

the corresponding contribution to the collision term $C[f]$ is given by

$$\begin{aligned}C[f_{a_1}](\vec{p}_1) = \frac{1}{2}\int\left[\prod_{i=2}^{m}\frac{d^3p_i}{(2\pi)^3 2E_i}\right]\left[\prod_{j=1}^{n}\frac{d^3p'_j}{(2\pi)^3 2E'_j}\right]\times(2\pi)^4\,\delta_D^{(4)}\left(\sum_{i=1}^{m}p_i - \sum_{j=1}^{n}p'_j\right)|\mathcal{M}|^2\\ \times\left[f_{b_1}\dots f_{b_n}\prod_{i=1}^{m}(1\pm f_{a_i}) - f_{a_1}f_{a_2}\dots f_{a_m}\prod_{j=1}^{n}(1\pm f_{b_j})\right],\end{aligned}$$

where $\mathcal{M}$ is the Feynman amplitude of the process, and we assumed crossing symmetry.

2.1.1 Boltzmann equation in Newtonian gauge

It is convenient to rewrite everything in terms of physical comoving variables $f = f(\tau, x^i, E, \hat{p}^i)$ in Newtonian gauge. Using formulas (1), the left-hand side of the Boltzmann equation (4.8) becomes (to first order in metric perturbation, but fully exact otherwise) check

$$\begin{aligned}\frac{df}{d\lambda} &= P^0\left[\frac{\partial f}{\partial\tau} + \frac{\partial f}{\partial x^i}\frac{P^i}{P^0} + \frac{\partial f}{\partial E}\frac{1}{P^0}\frac{dE}{d\lambda} + \frac{\partial f}{\partial\hat{p}^i}\frac{1}{P^0}\frac{d\hat{p}^i}{d\lambda}\right]\\ &= \frac{E}{a}\left[\frac{\partial f}{\partial\tau}(1-\psi) + \left(\hat{p}^j\partial_j f\right)\frac{p}{E}(1+\phi) - \frac{\partial f}{\partial E}\frac{p^2}{E}\left(\mathcal{H}(1-\psi)-\dot{\phi}\right) - \frac{\partial f}{\partial\bar{p}^j}\partial_j\psi\,E + \left(\frac{\partial f}{\partial p}\hat{p}^j - \frac{\partial f}{\partial\bar{p}^j}\right)\partial_j\phi\,\frac{p^2}{E}\right].\end{aligned} \tag{4.9}$$

In the formula above we included derivatives both versus $\vec{p}_j$ and the sole direction $\hat{p}_j$, since this writing will turn useful when taking moments.

2.2 Moments of the Boltzmann equation

The complete behavior of the species s in phase space is entirely described by the distribution f_s, which in turn is completely specified by the Boltzmann equation. In principle, we should solve the full system of Boltzmann equations, one for each species s, coupled through the mutual collision terms and the metric potentials, and Einstein equations, sourced by the energy-momentum of the species themselves. This task is extremely difficult, even numerically.

In perturbation theory, we can still make great progress by considering moments of the distribution function and the corresponding evolution equations obtained taking the same moments of the Boltzmann equation. In general, we define the bulk average of a quantity of interest Q for the fluid[4] by

$$\langle Q \rangle := \left(\int \frac{d^3p}{(2\pi)^3} f_s \right)^{-1} \int \frac{d^3p}{(2\pi)^3} \, Q \, f_s .$$

In particular, the covariant energy-momentum tensor is given by

$$T^{\mu}_{(s)\,\nu} = \int \frac{d^3P_i}{\sqrt{-g}(2\pi)^3} \, g_s \, \frac{P^\mu P_\nu}{2P^0} \, f_s = \int \frac{d^3\vec{p}}{(2\pi)^3} \, g_s \, \frac{P^\mu P_\nu}{2E} f_s .$$

In turn, number density, energy density and pressure are obtained as

$$n := \int \frac{d^3p}{(2\pi)^3} \, f_s , \quad \rho_s = T^0_0 = \int \frac{d^3p}{(2\pi)^3} E \, f_s , \quad P = \tfrac{1}{3} T^i_i = \int \frac{d^3\vec{p}}{(2\pi)^3} \, \frac{p^2}{3E} f_s .$$

Similarly the bulk velocity of the fluid, already introduced below equation (**??**), is obtained as

$$\vec{u} := \left\langle \frac{\vec{p}}{E} \right\rangle := \left(\int \frac{d^3p}{(2\pi)^3} f_s \right)^{-1} \int \frac{d^3p}{(2\pi)^3} \, \frac{\vec{p}}{E} \, f_s , \tag{4.10}$$

and the stress-tensor of the fluid is obtained as

$$n \, \sigma^{ij} := \left\langle \frac{p^i p^j}{E^2} \right\rangle - \left\langle \frac{p^i}{E} \right\rangle \left\langle \frac{p^j}{E} \right\rangle .$$

Taking moments of the Boltzmann equation generates the well-known Boltzmann hierarchy, where the evolution of a moment $\sim \left(\frac{\vec{p}}{E}\right)^n$ depends on higher moments $\sim \left(\frac{\vec{p}}{E}\right)^{n+1}$. In the next section we will consider *cold* matter. In this case $p \ll E \sim m$ and it will be sufficient to truncate this hierarchy at some order in $\frac{p}{E}$, typically $n = 2$. The treatment for radiation is far more delicate: in this case $p \sim E$, the above truncation argument is completely inadequate[5] and one has to account for the full hierarchy. This is a well-studied (and well-resolved) problem and we refer to pedagogical treatments in [Bau22, MB95, DS25].

2.2.1 Velocity averages for cold matter

Finally we clear a subtlety regarding statistical averages of powers $\frac{p_{i_1}}{E} \dots \frac{p_{i_n}}{E}$ for cold matter, in order to correctly identify perturbations of the same order in the next section. We will use the shorthand $v_i = \frac{p_i}{E}$ with modulus $v = \frac{p}{E}$. These quantities are not observable, they are just 'free' variables that need to be integrated against f, while we reserve the notation $\vec{u}$ for the (bulk) velocity of the fluid (4.10).

Following the notation in [MB95], the distribution function is split as

$$f(\tau, x, \vec{p}) =: \bar{f}(\tau, p) \, (1 + \Psi(\tau, x, \vec{p})) =: \bar{f}(\tau, p) + \delta f(\tau, x, \vec{p}),$$

where the zeroth-order term is independent of position $\vec{x}$ and direction $\hat{p}$ of momentum, due to isotropy and homogeneity of the background. The function Ψ, or equivalently δf, describes the small perturbations about the unperturbed solution.

[4] To simplify the notation, we do not indicate internal degrees of freedom. The quantities considered here are almost always independent of these. If this is the case, we implicitly include a suitable degeneracy factor g_s in the definition of f_s.

[5] In some cases, like for the photon-baryon fluid in the tight coupling regime this is actually still good.

From isotropy of the background, to order zero in field perturbations we have

$$\left\langle v_{i_1}\dots v_{i_n}\right\rangle := \left(\int\frac{d^3p}{(2\pi)^3}f\right)^{-1}\int\frac{d^3p}{(2\pi)^3}\,\frac{p}{E}\hat{p}_{i_1}\dots\frac{p}{E}\hat{p}_{i_n}\,f \overset{\text{order 0}}{\equiv} \begin{cases}\langle v^{2j}\rangle_0\left(\delta^{\text{Kr.}}_{i_1i_2}\dots\delta^{\text{Kr.}}_{i_{2j-1}i_{2j}}+\text{perm.}\right) & n=2j,\\ 0 & n=2j+1,\end{cases}$$

where $\langle\dots\rangle_0$ denotes averages taken with respect to the *background* distribution

$$\langle Q\rangle_0 := \left(\int\frac{d^3p}{(2\pi)^3}\bar{f}\right)^{-1}\int\frac{d^3p}{(2\pi)^3}\,Q\,\bar{f}.$$

Remark 2.1. The background distribution is centered around $\vec{v}=0$ with some thermal spread, say a Boltzmann distribution centered around zero with finite temperature. In the cold matter approximation $T_{\text{CDM}}\to 0$ we have a Dirac delta in momentum space

$$\bar{f}=\bar{n}(t)\,(2\pi)^3\delta^{(3)}(\vec{q}),$$

so that even all the norm averages $\langle v^\ell\rangle_0$ vanish. In any case, these background averages would be negligibly small, since they only arise from thermal spread around zero and redshift as $\langle v^n\rangle_0\propto a^{-n}$.

Perturbations to the distribution $\delta f(\tau,x,p,\hat{p})$ allow for nonisotropic averages $\left\langle v_{i_1}\dots v_{i_n}\right\rangle\neq 0$. Similarly to above, we could approximate f_s as the full space density $n_s(x,\tau)$ times some thermal spread around a *nonzero* peculiar velocity $\vec{u}_s(x,\tau)$, namely the bulk velocity of the fluid s. In the cold limit $T_{\text{CDM}}\to 0$ we now have

$$f_s=n_s(x,\tau)(2\pi)^3\delta^{(3)}(\vec{q}-m_s\vec{u}_s(x,\tau)),$$

so that all reduced moments vanish and we have

$$\left\langle v_{i_1}\dots v_{i_n}\right\rangle\simeq u_{i_1}\dots u_{i_n}.$$

Remark 2.2. In linear theory the peculiar velocity $u(x,t)\propto\dot{\delta}\propto\dot{D}_+=\mathcal{H}\,\mathbf{f}\,D_+$ grows in time (e.g. in EdS approximation $u\propto a^{1/2}$) and the full distribution f is centered around it in momentum space, regardless of the thermal spread being zero or not. Therefore, regardless of integrating over directions $\hat{v}$ and anisotropy, averages of $\frac{p}{E}$ are dominated by the peculiar velocity, in the sense that

$$\fint\frac{d^3q}{(2\pi)^3}\bar{f}\,v^\ell\propto T_{CDM}^{\frac{\ell}{2}}\propto a^{-\ell}\ll\fint\frac{d^3q}{(2\pi)^3}f\,v^\ell\sim\,|u(x,t)|^\ell\propto(\mathcal{H}\mathbf{f}D_+)^\ell\,.$$

In summary, we have two small parameters to expand with:

1. the perturbations δf, or better to the derive fields like δ_s and $\vec{u}_s$;
2. the thermal spread of matter characterized e.g. by the non-vanishing of *reduced* moments like $\langle v_iv_j\rangle-\langle v_i\rangle\langle v_j\rangle$, or equivalently, since the thermal spread of $\bar{f}$ and of f should be of the same order, by the magnitude of *background* averages $\langle\left(\frac{p}{E}\right)^\ell\rangle_0=\langle v^\ell\rangle_0$.

Following the standard EFT approach [**?**], we first expand in field perturbations and only then add back also the thermal spread with *effective* terms accounting for the microphysics we could not describe exactly.

Retaining powers $\langle\frac{p^2}{E^2}\rangle\sim u^2$ in our perturbative expansion is necessary because the contributions of a decay interaction to the RHS of the continuity and Euler equations beyond linear order are of the form $\Gamma\,\psi\,\delta_m$ and $\langle v^2\rangle\Gamma$ and these terms are essentially of the same order (cf. next section). Indeed, from the Poisson equation in the subhorizon limit $\delta_m\sim\frac{k^2}{\mathcal{H}^2}\psi$ and from the Euler equation (5.4) to leading order $\vec{u}\sim\frac{\vec{k}}{\mathcal{H}}\psi$, so that we have

$$\langle\frac{p^2}{E^2}\rangle\sim|u(x,t)|^2\sim\frac{k^2}{\mathcal{H}^2}\psi^2\sim\psi\,\delta_m.$$

In conclusion, we remark again that retaining powers $\langle\frac{p^2}{E^2}\rangle$ is not directly related to the order at which we truncate the Boltzmann hierarchy. Indeed, we will still close the continuity–Euler system by assuming no velocity dispersion

$$\frac{1}{n}\sigma_{ij}=\langle v_iv_j\rangle-\langle v_i\rangle\langle v_j\rangle\propto\text{ thermal spread }\simeq 0.$$

Only afterwards, during renormalization, will we introduce back effective terms like

$$\langle v_iv_j\rangle-\langle v_i\rangle\langle v_j\rangle=\delta_{ij}\,c_s^2\,\delta_m+\dots$$

In fact, in the same spirit, we might be forced to add back the corrections to $\Gamma\langle v^2\rangle=\Gamma u^2(1+\dots)$.

2.3 The continuity and Euler equations

We now take the zeroth and first moments of the of the Boltzmann equation (4.9) to obtain the continuity and Euler equations. After some tedious but elementary passages, the continuity equation reads (to first order in metric perturbations, fully exact otherwise)

$$\begin{aligned}\int \frac{d^3\vec{p}}{(2\pi)^3}\frac{1}{E}\frac{df}{d\lambda} &= \frac{1}{a}\left[(1-\psi)\frac{\partial n}{\partial\tau} + (1+\phi)\,\partial_j(n\,u_j) + 3\,n\left(\mathcal{H}(1-\psi)-\dot{\phi}\right) + n\,u_j\,(\partial_j\psi - 2\partial_j\phi)\right] \\ &= \int \frac{d^3\vec{p}}{(2\pi)^3}\frac{1}{E}C[f;f_{\text{other}}].\end{aligned} \tag{4.11}$$

In the absence of number-changing interactions, the RHS of the continuity equation vanishes. At order zero we find the usual Hubble dilution

$$\frac{\partial\bar{n}_\chi}{\partial\tau} + 3\,\bar{n}_\chi\mathcal{H} = a\left[\int\frac{d^3\vec{p}}{(2\pi)^3}\frac{1}{E}C[f]\right]_0$$

Ww then split the density as $n = \bar{n}(1+\delta)$ and get the perturbed continuity equation (to first order in metric perturbations, fully exact otherwise)

Do it

...

Similarly, integrating by parts and using (4.11) to simplify the $\partial_\tau n$ term, we obtain the Euler equation (to first order in metric, fully exact otherwise)

$$\begin{aligned}\int \frac{d^3\bar{p}}{(2\pi)^3}\frac{1}{E}\frac{df}{d\lambda}\frac{p\hat{p}_i}{E} = \frac{1}{a}\Big\{&(1-\psi)n\frac{\partial u_i}{\partial\tau} + (1+\phi)\,n\,u_j\,\partial_j u_i \\ &+ (1+\phi)\partial_j\Big[n\big(\langle v_i v_j\rangle - \langle v_i\rangle\langle v_j\rangle\big)\Big] \\ &+ \Big[\mathcal{H}(1-\psi)-\dot{\phi}\Big]\,n\,\big(\langle v_i\rangle - \langle v_i v^2\rangle\big) \\ &+ n\big[\delta^{ij} - \langle v_i\rangle\langle v_j\rangle\big]\,\partial_j\psi + n\big[\delta^{ij}\langle v^2\rangle - 3\langle v_i v_j\rangle + 2\langle v_i\rangle\langle v_j\rangle\big]\partial_j\phi \\ &+ a\,\langle v_i\rangle\int\frac{d^3\vec{p}}{(2\pi)^3}\frac{1}{E}C[f,f_{\text{other}}]\Big\} = \int\frac{d^3\vec{p}}{(2\pi)^3}\frac{1}{E}C[f,f_{\text{other}}]\frac{p\hat{p}_i}{E}.\end{aligned} \tag{4.12}$$

where we used the shorthand $v_i = \frac{p_i}{E}$. Note that the last term $a\,\langle v_i\rangle\int\frac{d^3\vec{p}}{(2\pi)^3}\frac{1}{E}C[f,f_{\text{other}}]$ at the LHS appeared since we used (4.11) to simplify the $\partial_\tau n$ term. It will cancel up to some order in $\langle(\frac{p}{E})^\ell\rangle$ with corresponding terms at the RHS when expanding in powers $\langle\frac{p}{E}\rangle$ for cold matter. We remark that $\langle v_i\rangle \equiv u_i$ exactly by definition of bulk velocity (4.10), while the identification $\langle v_i v_j\rangle \simeq u_i u_j$ holds with the caveats of Section 2.2.1. Multiplying by $1+\psi$, dropping second orders in metric perturbations and third orders $\sim\langle v\rangle^3$ and $\sim\phi\langle v\rangle^2$, and dividing by the density n yield the familiar Euler equation for cold matter (now correct to second order in $\langle v\rangle$, and to first order in metric perturbations)

$$\frac{\partial u_i}{\partial\tau} + u_j\,\partial_j u_i + \frac{1}{n}\partial_j\Big[n\big(\langle v_i v_j\rangle - \langle v_i\rangle\langle v_j\rangle\big)\Big] + \Big[\mathcal{H}-\dot{\phi}\Big]\,u_i + \partial_i\psi = (\text{interactions}) + O\big(\langle v^3\rangle\big).$$

In conclusion, we drop leftover order $\geqslant 3$ terms and separate linear and quadratic terms in the fields to get the final versions of the continuity and Euler equation (correct to second order in perturbations, upon ignoring terms $\sim\phi^2$):

Fix from here below

$$\begin{aligned}\dot{\delta}_\chi + \partial_j u^j_\chi - 3\dot{\phi} + \psi\,a\,\Gamma &= -\partial_j\left(\delta_\chi\,u^j_\chi\right) + 3\dot{\phi}\delta_\chi - (\phi+\psi)\partial_j u^j_\chi - u^j_\chi\,(\partial_j\psi - 2\partial_j\phi) - \psi\,\delta_\chi\,a\,\Gamma + \frac{1}{2}\langle v^2_\chi\rangle\,a\,\Gamma, \\ \dot{u}^i_\chi + \mathcal{H}u^i_\chi + \partial_i\psi &= -\,u^j_\chi\,\partial_j u^i_\chi - \frac{1}{n_\chi}\partial_j\Big[n_\chi\big(\langle v^i_\chi v^j_\chi\rangle - \langle v^i_\chi\rangle\langle v^j_\chi\rangle\big)\Big] + \dot{\phi}\,u^i_\chi.\end{aligned}$$

Taking the divergence of the Euler equation, with the usual notation $\nabla\cdot u =: \theta$, finally yields

$$\dot{\theta}_\chi + \mathcal{H}\,\theta_\chi + \nabla^2\psi = -\,\partial_i\left(u^j_\chi\,\partial_j u^i_\chi\right) - \frac{1}{n_\chi}\,\partial_i\partial_j\Big[n_\chi\big(\langle v^i_\chi v^j_\chi\rangle - \langle v^i_\chi\rangle\langle v^j_\chi\rangle\big)\Big] + \partial_i\left(\dot{\phi}\,u^i_\chi\right).$$

Remark 2.3. We comment on the color coding in the above equations, a full analysis is postoponed to Section **??**. The black parts are the usual equations for higher order perturbation theory in a ΛCDM universe. The term $3\dot{\phi}$ is a linear term already present in a ΛCDM universe, but typically dropped in the subhorizon limit. The decay will somewhat enhance the growth over time of this term and reduce its subhorizon suppression. The term $\psi\, a\, \Gamma$ is the linear correction from dcdm. We will argue that it is of the same order of $3\dot{\phi}$. Thus, if we want to see explicit linear corrections from dcdm beyond the pure background effect, we need to keep both terms and cannot invoke the subhorizon limit completely.

The situation is analogous at second order. The green terms $\sim \phi\, \partial_j u$ are quadratic terms already present in a ΛCDM universe, again typically dropped since metric perturbations are much smaller than matter ones. These are genuinely suppressed in the subhorizon limit, regardless of the presence of decay. The red terms $\sim \dot{\phi}\, \delta\,,\ \dot{\phi}\, u$ are other quadratic terms already present in a ΛCDM universe and again typically dropped in the subhorizon limit. As for the orange term above, the decay will enhance the growth over time of these terms and reduce the suppression. Finally the blue terms are the quadratic corrections from dcdm. Similarly to above, we will argue that these quadratic term from dcdm are of the same order as the red ones. Thus if we want to see genuine quadratic corrections from dcdm, we must to keep both types of terms.

Finally we have the velocity dispersion term. Neglecting this term does not depend on the subhorizon limit, rather it is small due to the thermal spread of cold matter. We already discussed that we will initially take this term to be zero, and only then add back effective corrections accounting for the microphysics we could not capture exactly.

Finally we discuss baryons and 'standard' non-decaying CDM. If we consider times well after recombination so that baryon perturbations have already caught up with CDM ones, we can treat baryons and stable CDM as a single fluid (stable matter, denoted s). The only difference from the above analysis for dcdm is the absence of the decay term. In this case, the perturbed continuity and Euler equations are simply (correct to second order in perturbations, upon ignoring terms $\sim \phi^2$)

$$\begin{aligned}
\dot{\delta}_s + \partial_j u_s^j - 3\dot{\phi} &= -\partial_j\big(\delta_s\, u_s^j\big) + 3\dot{\phi}\delta_s - (\phi+\psi)\partial_j u_s^j - u_s^j\,(\partial_j\psi - 2\partial_j\phi),\\
\dot{u}_s^i + \mathcal{H} u_s^i + \partial_i\psi &= -\,u_s^j\,\partial_j u_s^i - \frac{1}{n_s}\partial_j\Big[n_s\big(\langle v_s^i v_s^j\rangle - \langle v_s^i\rangle\langle v_s^j\rangle\big)\Big] + \dot{\phi}\, u_s^i,
\end{aligned}$$

with the same considerations on the colored terms as above.

3 Collision term and RHS of Boltzmann equation

An excellent treatment of how to deal with the collision term when taking moments of the Boltzmann equation is given in [BAES18].

4 Energy-momentum tensor & Bianchi identities

The complete behavior of the species s is described by the distribution f_s. In particular, the energy-momentum tensor is given by

$$T^{\mu}_{(s)\,\nu} = \int \frac{d^3P_i}{\sqrt{-g}(2\pi)^3}\,\frac{P^\mu P_\nu}{P^0}\, f_s = \int \frac{d^3\vec{p}}{(2\pi)^3}\,\frac{P^\mu P_\nu}{E}\, f_s.$$

The form of the energy-momentum tensor is justified since

Finish or move to GR

Using formulas (4.5) we compute

$$T^0_0 = \int \frac{d^3\vec{p}}{(2\pi)^3} E\, f_s, \quad T^0_i = \int \frac{d^3\vec{p}}{(2\pi)^3}(1-\phi-\psi)\, p\hat{p}^i\, f_s, \quad T^i_j = \int \frac{d^3\vec{p}}{(2\pi)^3}\,\frac{p^2\hat{p}^i\hat{p}^j}{E}\, f_s.$$

We now insert an equilibrium distribution with local temperature

Write the general case (cf. decaying)

$$f(\tau, x^i, E, \hat{p}^i) = f(\tfrac{E}{T}), \quad T = \bar{T}(\tau)(1+\Theta(\tau, x^i, p, \hat{p}^i)) = \bar{T} + \delta T \quad \Rightarrow \quad f \simeq \bar{f} - \frac{\partial \bar{f}}{\partial E} E\Theta + O(2).$$

In the case of photons, taking $E = p$ and assuming Θ independent of E for simplicity we get Then we compute

Finish

$$
\begin{aligned}
T^0_0 &= -\int \frac{d^3\vec{p}}{(2\pi)^3} E\,\bar{f} + \int \frac{d^3\vec{p}}{(2\pi)^3} E^2 \frac{\partial \bar{f}}{\partial E}\Theta = -\bar{\rho} - \bar{\rho}\Theta_0, \\
T^0_i &= \int \frac{d^3\vec{p}}{(2\pi)^3}(1-\phi-\psi)\,p\hat{p}^i\,\bar{f} - \int \frac{d^3\vec{p}}{(2\pi)^3}(1-\phi-\psi)\,p\hat{p}^i\,E\frac{\partial \bar{f}}{\partial E}\Theta = -4\bar{\rho} i\hat{k}^i\Theta_1 = -\tfrac{1}{3}(\bar{\rho}+\bar{p})i\hat{k}^i\Theta_1, \\
T^i_j &= \int \frac{d^3\vec{p}}{(2\pi)^3}\frac{p^2\hat{p}^i\hat{p}^j}{E}\bar{f} - \int \frac{d^3\vec{p}}{(2\pi)^3}\frac{p^2\hat{p}^i\hat{p}^j}{E}E\frac{\partial \bar{f}}{\partial E}\Theta.
\end{aligned}
$$

Comparing with the perturbed perfect fluid energy-momentum tensor

$$
\begin{aligned}
&T^\mu_\nu = u^\mu u_\nu(\rho + P) + P\delta^\mu_\nu + \Sigma^\mu_\nu \\
&T^0_0 = -\bar{\rho} - \delta\rho, \quad T^0_i = -(\bar{\rho}+\bar{P})v_i, \quad T^i_j = (\bar{P}+\delta P)\delta^i_j + \Sigma^i_j
\end{aligned}
$$

we find

$$
\delta_\gamma = \frac{\delta\rho}{\bar{\rho}} = 4\Theta_0, \quad v_i = -\frac{3}{4}\int \frac{d^3\vec{p}}{(2\pi)^3} p\hat{p}_i E\frac{\partial \bar{f}}{\partial E}\Theta, \quad \Sigma^i_j = -\int \frac{d^3\vec{p}}{(2\pi)^3}\frac{p^2\hat{p}^i\hat{p}^j}{E}E\frac{\partial \bar{f}}{\partial E}\Theta.
$$

It is convenient to isolate the 2 *scalar* degrees of freedom of the (i,j)-component of the stress-tensor and split them into trace and traceless part. Namely we project[6] the scalar part using

$$
P^i_j := \tfrac{k^ik_j}{k^2} - \tfrac{1}{3}\delta^{ij}, \;\; \delta^i_j P^j_i = 0, \;\; P^i_j P^j_i = \tfrac{2}{3} \;\Rightarrow\; (T^i_j)_{\text{scalar}} = \tfrac{1}{3}\delta^i_j\,T^k_k + \left(\tfrac{k^ik_j}{k^2} - \tfrac{1}{3}\delta^{ij}\right)\left[\tfrac{3}{2}\left(\tfrac{k^bk_a}{k^2} - \tfrac{1}{3}\delta^b_a\right)T^a_b\right]
$$

Isotropy of the backround implies

$$
\int \frac{d^3\vec{p}}{(2\pi)^3}\frac{p^2\hat{p}^i\hat{p}^j}{E}\bar{f} = \int \frac{d^3\vec{p}}{(2\pi)^3}\frac{p^2}{E}\tfrac{1}{3}\delta^i_j\bar{f} = \delta^i_j\bar{P}
$$

Explicitly we have

$$
\tfrac{1}{3}d^j_iT^i_j = \int \frac{d^3\vec{p}}{(2\pi)^3}\frac{p^2}{3E}\left(\bar{f} - \Theta E\frac{\partial \bar{f}}{\partial E}\right) = \bar{P} + \int \frac{d^3\vec{p}}{(2\pi)^3}\frac{p^2}{3}\Theta\frac{\partial \bar{f}}{\partial E} = \bar{P} + \tfrac{4}{3}\bar{\rho}\Theta_0 = \bar{P} + \delta P \;\Rightarrow \delta P = \tfrac{4}{3}\bar{\rho}\Theta_0 = \tfrac{1}{3}\bar{\rho}\delta_\gamma.
$$

As confirmed by $\int d\mu(\mu^2 - \frac{1}{3}) = 0$ the traceless part of the background vanishes and integrating by parts we get

$$
\begin{aligned}
\left(\tfrac{k^jk_i}{k^2} - \tfrac{1}{3}\delta^j_i\right)T^i_j &= \left(\tfrac{k^jk_i}{k^2} - \tfrac{1}{3}\delta^j_i\right)\int \frac{d^3\vec{p}}{(2\pi)^3}\frac{p^2\hat{p}^i\hat{p}^j}{E}f = \int \frac{d^3\vec{p}}{(2\pi)^3}\frac{p^2}{E}(\mu^2 - \tfrac{1}{3})\left(\bar{f} - \Theta E\frac{\partial \bar{f}}{\partial E}\right) \\
&= 4\int \frac{d^3\vec{p}}{(2\pi)^3}\,p\bar{f}\frac{1}{(4\pi)}\int d\Omega\,\tfrac{2}{3}\left(\tfrac{3\mu^2-1}{2}\right)\Theta = 4\bar{\rho} = 4\bar{\rho}\left(-\tfrac{2}{3}\Theta_2\right).
\end{aligned}
$$

In conclusion we have

$$
(T^i_j)_{\text{scalar}} = \delta^i_j\left(\bar{P} + \tfrac{4}{3}\bar{\rho}\Theta_0\right) - 4\left(\tfrac{k^ik_j}{k^2} - \tfrac{1}{3}\delta^{ij}\right)\bar{\rho}\Theta_2.
$$

5 Einstein–Boltzmann system

The Einstein equation for $\psi - \phi \propto k^{-2}\Theta_2$ seems to suggest anisotropic stress is relevant for making $\psi \neq \phi$ only on large scales since it is suppressed by k^{-2} and that it is also relevant at ealry times since then radiation dies off as a^{-4} and matter grows as a^{-3}, so that the ratio of radiation to matter energy density is larger at early times. In particular anisotropic stress may lead $\psi \neq \phi$ only due to anisotropic stress of neutrinos and then around recombination due to anisotropic stress of photons that develop a nonnegligible quadrupole moment Θ_2.

do it

6 Higher-order perturbation theory

[6] Note P^i_j is not really a projector in the sense of $\mathbb{R}^3$ but rather in the SVT decomposition sense.

5 Dark Matter & Baryons

1 Equation of motion & phase space distribution

We report the newtonian-gauge geodesic equation (4.6)-(1)

$$\begin{aligned}
\frac{dE}{d\tau} &= -\frac{p^2}{E}(\mathcal{H}-\dot{\phi}) - p\hat{p}^j\partial_j\psi, \\
\frac{dp}{d\tau} &= -p(\mathcal{H}-\dot{\phi}) - E\hat{p}^j\partial_j\psi, \\
\frac{d\hat{p}^i}{d\tau} &= \left(\hat{p}^i\hat{p}^j - \delta^{ij}\right)\left(\frac{E}{p}\partial_j\psi + \frac{p}{E}\partial_j\phi\right).
\end{aligned}$$

At the background level, the second one integrates to $p \propto a^{-1}$, giving the usual redshift of momenta. In turn the first one integrates to dispersion relation

$$\frac{dE^2}{d\tau} = -2p^2\mathcal{H} = \frac{d}{d\tau}a^{-2} \quad \Rightarrow \quad E^2 = m^2 + \frac{p_0^2}{a^2}.$$

The Boltzmann equation (4.9) is also reported (to second order in metric, exact otherwise)

$$\begin{aligned}
C[f] = \frac{df}{d\lambda} &= P^0\left[\frac{\partial f}{\partial\tau} + \frac{\partial f}{\partial x^i}\frac{P^i}{P^0} + \frac{\partial f}{\partial E}\frac{1}{P^0}\frac{dE}{d\lambda} + \frac{\partial f}{\partial\hat{p}^i}\frac{1}{P^0}\frac{d\hat{p}^i}{d\lambda}\right] \\
&= \frac{E}{a}\left[\frac{\partial f}{\partial\tau}(1-\psi) + \left(\hat{p}^j\partial_j f\right)\frac{p}{E}(1+\phi) - \frac{\partial f}{\partial E}\frac{p^2}{E}\left(\mathcal{H}(1-\psi)-\dot{\phi}\right) - \frac{\partial f}{\partial\bar{p}^j}\partial_j\psi\, E + \left(\frac{\partial f}{\partial p}\hat{p}^j - \frac{\partial f}{\partial\bar{p}^j}\right)\partial_j\phi\,\frac{p^2}{E}\right].
\end{aligned} \tag{5.1}$$

We can now take moments and expand in powers of p/m....

cf. my thesis & whint L11 & dodelson 12

1.1 The continuity and Euler equations

We now take the zeroth and first moments of the LHS of the Boltzmann equation (5.1) to obtain the LHS of the continuity and Euler equations. After some tedious but elementary passages, the continuity equation reads (to first order in metric perturbations, fully exact otherwise)

$$\int\frac{d^3\vec{p}}{(2\pi)^3}\frac{1}{E}\frac{df}{d\lambda} = \frac{1}{a}\left[(1-\psi)\frac{\partial n}{\partial\tau} + (1+\phi)\,\partial_j(n\,u_j) + 3\,n\left(\mathcal{H}(1-\psi)-\dot{\phi}\right) + n\,u_j\left(\partial_j\psi - 2\partial_j\phi\right)\right].$$

Combining with the RHS from (**??**) the continuity equation reads (again to first order in metric perturbations, fully exact otherwise)

$$\frac{\partial n}{\partial\tau} + \left(1+\phi+\psi\right)\partial_j(n\,u_j) + 3\,n\left(\mathcal{H}-\dot{\phi}\right) + n\,u_j\left(\partial_j\psi - 2\partial_j\phi\right) = -a\,n\langle\Gamma\rangle(1+\psi). \tag{5.2}$$

Similarly, integrating by parts and using (5.2) to simplify the $\partial_\tau n$ term, we obtain the LHS of the Euler equation (to first order in metric, fully exact otherwise)

$$\begin{aligned}
\int\frac{d^3\bar{p}}{(2\pi)^3}\frac{1}{E}\frac{df}{d\lambda}\frac{p\hat{p}_i}{E} = \frac{1}{a}\Bigg\{&(1-\psi)n\frac{\partial u_i}{\partial\tau} + (1+\phi)\,n\,u_j\,\partial_j u_i \\
&+ (1+\phi)\partial_j\Big[n\big(\langle v_iv_j\rangle - \langle v_i\rangle\langle v_j\rangle\big)\Big] \\
&+ \Big[\mathcal{H}(1-\psi)-\dot{\phi}\Big]n\,\left(\langle v_i\rangle - \langle v_iv^2\rangle\right) \\
&+ n\big[\delta^{ij} - \langle v_i\rangle\langle v_j\rangle\big]\,\partial_j\psi + n\big[\delta^{ij}\langle v^2\rangle - 3\langle v_iv_j\rangle + 2\langle v_i\rangle\langle v_j\rangle\big]\partial_j\phi \\
&- a\,\langle v_i\rangle\,n\,\langle\Gamma\rangle\Bigg\},
\end{aligned} \tag{5.3}$$

where we used the shorthand $v_i = \frac{p_i}{E}$. Note that the last term $-a\langle v_i\rangle n\langle\Gamma\rangle$ appeared since we used (**??**) to simplify the $\partial_\tau n$ term and it will cancel with the corresponding RHS up to second order in $\langle\frac{p}{E}\rangle$ included. Namely, equating (**??**) and (5.3), we have

$$
\begin{aligned}
\cancel{-a\,n\langle\vec{v}\rangle\Gamma\Big(1-\frac{1}{2}\langle v^2\rangle\Big)} + O\big(\langle v^3\rangle\big) = & \cancel{-a\langle v_i\rangle n\langle\Gamma\rangle} \\
& + (1-\psi)\,n\,\frac{\partial u_i}{\partial\tau} + (1+\phi)\,n\,u_j\,\partial_j u_i \\
& + (1+\phi)\;\partial_j\Big[n\,\big(\langle v_i v_j\rangle - \langle v_i\rangle\langle v_j\rangle\big)\Big] \\
& + \Big[\mathcal{H}(1-\psi) - \dot{\phi}\Big]\,n\;\big(\langle v_i\rangle - \langle v_i v^2\rangle\big) \\
& + n\Big(\delta^{ij} - \langle v_i\rangle\langle v_j\rangle\Big)\partial_j\psi + n\Big(\delta^{ij}\langle v^2\rangle - 3\,\langle v_i v_j\rangle + 2\langle v_i\rangle\langle v_j\rangle\Big)\partial_j\phi.
\end{aligned}
\tag{5.4}
$$

Finally, multiplying by $1+\psi$, dropping second orders in metric perturbations and third orders $\sim\langle v\rangle^3$ and $\sim\phi\langle v\rangle^2$, and dividing by the density n yield the Euler equation (now correct to second order in $\langle v\rangle$, and to first order in metric perturbations)

$$
\frac{\partial u_i}{\partial\tau} + u_j\,\partial_j u_i + \frac{1}{n}\partial_j\Big[n\big(\langle v_i v_j\rangle - \langle v_i\rangle\langle v_j\rangle\big)\Big] + \Big[\mathcal{H} - \dot{\phi}\Big]\,u_i + \partial_i\psi = 0 + O\big(\langle v^3\rangle\big).
$$

We remark that $\langle v_i\rangle \equiv u_i$ exactly by definition of bulk velocity (4.10), while the identification $\langle v_i v_j\rangle \simeq u_i u_j$ holds with the caveats of Section 2.2.1.

We now split the density $n = \bar{n}(1+\delta)$ and in turn split (**??**) into background and perturbation part. As we anticipated in (**??**), the zeroth order continuity equation is

$$
\frac{\partial\bar{n}_\chi}{\partial\tau} + 3\,\bar{n}_\chi\mathcal{H} = -a\,\bar{n}_\chi\bar{\Gamma}.
$$

Using the expansion (**??**), the perturbed continuity equation reads (now correct to second order in $\langle v\rangle$, and to first order in metric perturbations)

$$
\dot{\delta}_\chi + \big(1+\phi+\psi\big)\,\partial_j\big[(1+\delta)\,u^j_\chi\big] - 3\dot{\phi}(1+\delta_\chi) + u^j_\chi\,(\partial_j\psi - 2\partial_j\phi) = -\psi(1+\delta_\chi)\,a\,\Gamma + \frac{1}{2}\langle v^2_\chi\rangle\,a\,\Gamma.
$$

In conclusion, we drop leftover order $\geqslant 3$ terms and separate linear and quadratic terms in the fields to get the final versions of the continuity and Euler equation (correct to second order in perturbations, upon ignoring terms $\sim\phi^2$):

$$
\begin{aligned}
\dot{\delta}_\chi + \partial_j u^j_\chi - 3\dot{\phi} + \psi\,a\,\Gamma &= -\partial_j\big(\delta_\chi\,u^j_\chi\big) + 3\dot{\phi}\delta_\chi - (\phi+\psi)\partial_j u^j_\chi - u^j_\chi\,(\partial_j\psi - 2\partial_j\phi) - \psi\,\delta_\chi\,a\,\Gamma + \frac{1}{2}\langle v^2_\chi\rangle\,a\,\Gamma, \\
\dot{u}^i_\chi + \mathcal{H}u^i_\chi + \partial_i\psi &= -\,u^j_\chi\,\partial_j u^i_\chi - \frac{1}{n_\chi}\partial_j\Big[n_\chi\big(\langle v^i_\chi v^j_\chi\rangle - \langle v^i_\chi\rangle\langle v^j_\chi\rangle\big)\Big] + \dot{\phi}\,u^i_\chi.
\end{aligned}
$$

Taking the divergence of the Euler equation, with the usual notation $\nabla\cdot u =: \theta$, finally yields

$$
\dot{\theta}_\chi + \mathcal{H}\,\theta_\chi + \nabla^2\psi = -\,\partial_i\big(u^j_\chi\,\partial_j u^i_\chi\big) - \frac{1}{n_\chi}\,\partial_i\partial_j\Big[n_\chi\big(\langle v^i_\chi v^j_\chi\rangle - \langle v^i_\chi\rangle\langle v^j_\chi\rangle\big)\Big] + \partial_i\left(\dot{\phi}\,u^i_\chi\right).
$$

Remark 1.1. We comment on the color coding in the above equations, a full analysis is postoponed to Section **??**. The black parts are the usual equations for higher order perturbation theory in a ΛCDM universe. The term $3\dot{\phi}$ is a linear term already present in a ΛCDM universe, but typically dropped in the subhorizon limit. The decay will somewhat enhance the growth over time of this term and reduce its subhorizon suppression. The term $\psi\,a\,\Gamma$ is the linear correction from dcdm. We will argue that it is of the same order of $3\dot{\phi}$. Thus, if we want to see explicit linear corrections from dcdm beyond the pure background effect, we need to keep both terms and cannot invoke the subhorizon limit completely.

The situation is analogous at second order. The green terms $\sim\phi\,\partial_j u$ are quadratic terms already present in a ΛCDM universe, again typically dropped since metric perturbations are much smaller than matter ones. These

are genuinely suppressed in the subhorizon limit, regardless of the presence of decay. The red terms $\sim \dot{\phi}\,\delta\,,\ \dot{\phi}\,u$ are other quadratic terms already present in a ΛCDM universe and again typically dropped in the subhorizon limit. As for the orange term above, the decay will enhance the growth over time of these terms and reduce the suppression. Finally the blue terms are the quadratic corrections from dcdm. Similarly to above, we will argue that these quadratic term from dcdm are of the same order as the red ones. Thus if we want to see genuine quadratic corrections from dcdm, we must to keep both types of terms.

Finally we have the velocity dispersion term. Neglecting this term does not depend on the subhorizon limit, rather it is small due to the thermal spread of cold matter. We already discussed that we will initially take this term to be zero, and only then add back effective corrections accounting for the microphysics we could not capture exactly.

Finally we discuss baryons and 'standard' non-decaying CDM. If we consider times well after recombination so that baryon perturbations have already caught up with CDM ones, we can treat baryons and stable CDM as a single fluid (stable matter, denoted s). The only difference from the above analysis for dcdm is the absence of the decay term. In this case, the perturbed continuity and Euler equations are simply (correct to second order in perturbations, upon ignoring terms $\sim \phi^2$)

$$\dot{\delta}_s + \partial_j u_s^j - 3\dot{\phi} = -\partial_j\left(\delta_s\, u_s^j\right) + 3\dot{\phi}\delta_s - (\phi+\psi)\partial_j u_s^j - u_s^j\left(\partial_j\psi - 2\partial_j\phi\right),$$
$$\dot{u}_s^i + \mathcal{H}u_s^i + \partial_i\psi = -\,u_s^j\,\partial_j u_s^i - \frac{1}{n_s}\partial_j\Big[n_s\big(\langle v_s^i v_s^j\rangle - \langle v_s^i\rangle\langle v_s^j\rangle\big)\Big] + \dot{\phi}\,u_s^i,$$

with the same considerations on the colored terms as above.

1.2 Baryons

2 Dark Matter Production

There are various mechanisms to produce dark matter in the early universe. We can broadly classify them into thermal and non-thermal mechanisms as follows: Redo

- **Thermal mechanisms**: In these scenarios, dark matter particles were once in thermal equilibrium with the Standard Model particles in the early universe. As the universe expanded and cooled, the interaction rates dropped below the Hubble expansion rate, leading to a "freeze-out" of dark matter particles. The relic abundance of dark matter is determined by the annihilation cross-section and mass of the dark matter particles. A classic example of this mechanism is the Weakly Interacting Massive Particle (WIMP) paradigm.

- **Non-thermal mechanisms**: In these scenarios, dark matter particles were never in thermal equilibrium with the Standard Model particles. Instead, they were produced through processes such as the decay of heavier particles, phase transitions, or other non-equilibrium processes. The relic abundance in these cases depends on the specifics of the production mechanism rather than thermal freeze-out. An example of this mechanism is the production of axions through the misalignment mechanism.

Back of the envelope estimates for DM production
Even without the full machinery of Boltzmann equations, we can get rough estimates for the relic abundance of dark matter produced via thermal and non-thermal mechanisms by considering the relevant interaction rates and the expansion rate of the universe.

2.1 Thermal mechanisms

2.1.1 Freeze-out

2.2 Non-thermal mechanisms

2.2.1 Freeze-in

6 Cosmic Microwave Background

1 Geodesic & Boltzmann Equations for photons

We report the newtonian-gauge geodesic equations (4.6)-(1) fot photons, which simplify since $E = p$,

$$\begin{aligned}
\frac{dP^0}{d\lambda} &= -\frac{E^2}{a^2}\left(\mathcal{H}(1-2\psi)+\dot{\psi}\right) - 2\frac{Ep}{a^2}\hat{p}^j\partial_j\psi - \frac{p^2}{a^2}\left(\mathcal{H}(1-2\psi)+\dot{\phi}\right),\\
\frac{dP^k}{d\lambda} &= -\frac{E^2}{a^2}\partial_k\psi - \frac{Ep}{a^2}\hat{p}^k\, 2\left(\mathcal{H}(1-\psi+\phi)-\dot{\phi}\right) - \frac{p^2}{a^2}\left(\partial_k\phi - 2\hat{p}^k\hat{p}^j\partial_j\phi\right).\\
\frac{dE}{d\tau} &= -\frac{E^2}{p}(\mathcal{H}-\dot{\phi}) - p\hat{p}^j\partial_j\psi = -E\mathcal{H} + E\frac{d\phi}{d\tau} - p\hat{p}^j(\partial_j\phi+\partial_j\psi),\\
\frac{d\hat{p}^i}{d\tau} &= \left(\hat{p}^i\hat{p}^j - \delta^{ij}\right)\left(\frac{E}{p}\partial_j\psi + \frac{p}{E}\partial_j\phi\right).
\end{aligned}$$

The last equation implies that to zeroth order in metric perturbations, photons travel along straight lines, while at first order they are deflected by the gravitational potentials ψ and ϕ. The third equation encodes the change in energy along the trajectory. It is convenient to rewrite it trading partial with total time derivatives $\frac{\partial\phi}{\partial\tau} = \frac{d\phi}{d\tau} - \frac{\partial\phi}{\partial x^j}\frac{dx^j}{d\tau}$ to get

$$\frac{d\log E}{d\tau} = -\mathcal{H} + \frac{d\phi}{d\tau} - \hat{p}^j(\partial_j\phi+\partial_j\psi).$$

To order zero it encode the expected redshift behavior $E = p = \propto a^{-1}$. The perturbed part implies energy is red-blueshifted by the gravitational potentials via the istantaneous and integrated Sachs-Wolf effect. Since the change in $\hat{p}^j$ along the trajectory is already first order, we can integrate it to get

$$\begin{aligned}
\log\left(\frac{E_f}{E_i}\right) &= \int_{\tau_i}^{\tau_f} d\tau\ \ -\mathcal{H} + \frac{d\phi}{d\tau} - \hat{p}^j(\partial_j\phi+\partial_j\psi)\, d\tau\\
&= \log\left(\frac{a_i}{a_f}\right) + \underbrace{\left(\phi(\tau_f,x_f)-\phi(\tau_i,x_i)\right)}_{\text{instantaneous SW (aka gravitational redshift)}} - \int_{\tau_i}^{\tau_f}\underbrace{d\tau\ \hat{p}^j(\partial_j\phi+\partial_j\psi)}_{\text{integrated Sachs-Wolfe effect}}.
\end{aligned}$$

Writing $E = \bar{E}(1+\Theta)$ with $\bar{E}\propto a^{-1}$ so that $\bar{E}_i a_i = \bar{E}_f a_f$ to remove the background redshift, we get the expected Sachs-Wolfe formula for the observed temperature anisotropies

$$\Theta_f(\hat{p}) - \Theta_i(\hat{p}) \simeq \log\left(\frac{1+\Theta_f(\hat{p})}{1+\Theta_i(\hat{p})}\right) = \phi(\tau_f,x_f) - \phi(\tau_i,x_i) - \int_{\tau_i}^{\tau_f} d\tau\ \hat{p}^j(\partial_j\phi+\partial_j\psi).$$

We report the Boltzmann equation (4.9) for photons becomes (second order in metric perturbations, exact otherwise)

$$\begin{aligned}
C[f] &= \frac{df}{d\lambda} = P^0\left[\frac{\partial f}{\partial\tau} + \frac{\partial f}{\partial x^i}\frac{P^i}{P^0} + \frac{\partial f}{\partial E}\frac{1}{P^0}\frac{dE}{d\lambda} + \frac{\partial f}{\partial\hat{p}^i}\frac{1}{P^0}\frac{d\hat{p}^i}{d\lambda}\right]\\
&= \frac{E}{a}\left[\frac{\partial f}{\partial\tau}(1-\psi) + \left(\hat{p}^j\partial_j f\right)\frac{p}{E}(1+\phi) - \frac{\partial f}{\partial E}\frac{p^2}{E}\left(\mathcal{H}(1-\psi)-\dot{\phi}\right) - \frac{\partial f}{\partial\bar{p}^j}\partial_j\psi\, E + \left(\frac{\partial f}{\partial p}\hat{p}^j - \frac{\partial f}{\partial\bar{p}^j}\right)\partial_j\phi\,\frac{p^2}{E}\right].
\end{aligned} \tag{6.1}$$

Photons retain local equilibrium with a blackbody distribution at all time, simply with a time and momentum dependent temperature, Discuss

$$f(\tau,x^i,E,\hat{p}^i) = \frac{1}{\exp\left(\frac{E}{T(\tau,x^i)}\right)-1}, \quad T = \bar{T}(\tau)(1+\Theta(\tau,x^i,p,\hat{p}^i)) = \bar{T} + \delta T.$$

More generally, assuming the full distribution is still a function $f(E/T)$ of the ratio E/T to some *local* temperature T, we Taylor expand around the background $\bar{f}(\tau,E) = \bar{f}(E/\bar{T}(\tau))$ to get

$$f = \bar{f} + \frac{\partial f}{\partial T}\delta T = \bar{f} - \frac{\partial\bar{f}}{\partial E}E\Theta = \bar{f} - \frac{d\bar{f}}{d(E/\bar{T})}\frac{E}{\bar{T}}\Theta, \qquad \text{with} \quad \frac{d\bar{f}}{d(E/\bar{T})} = \frac{\partial\bar{f}}{\partial E}_{|\tau}\bar{T} = -\frac{\partial\bar{f}}{\partial\bar{T}}_{|E}\frac{\bar{T}^2}{E}.$$

In turn we compute the various derivatives appearing in the LHS of Boltzmann equation

$$\begin{aligned}
\frac{\partial f}{\partial \tau} &= -\frac{d\bar{f}}{d(E/\bar{T})}\frac{E}{\bar{T}^2}\frac{d\bar{T}}{d\tau} + \frac{d^2\bar{f}}{d(E/\bar{T})^2}\frac{E^2}{\bar{T}^3}\frac{d\bar{T}}{d\tau}\Theta + \frac{d\bar{f}}{d(E/\bar{T})}\frac{E}{\bar{T}^2}\frac{d\bar{T}}{d\tau}\Theta - \frac{d\bar{f}}{d(E/\bar{T})}\frac{E}{\bar{T}}\frac{\partial\Theta}{\partial\tau} \\
&= \frac{\partial\bar{f}}{\partial E}\frac{E}{\bar{T}}\frac{d\bar{T}}{d\tau} + \frac{\partial^2\bar{f}}{\partial E^2}\frac{E^2}{\bar{T}}\frac{d\bar{T}}{d\tau}\Theta + \frac{\partial\bar{f}}{\partial E}\frac{E}{\bar{T}}\frac{d\bar{T}}{d\tau}\Theta - \frac{\partial\bar{f}}{\partial E}E\frac{\partial\Theta}{\partial\tau}, \\
\frac{\partial f}{\partial E} &= \frac{\partial\bar{f}}{\partial E} - \frac{\partial^2\bar{f}}{\partial E^2}E\Theta - \frac{\partial\bar{f}}{\partial E}\Theta - \frac{\partial\bar{f}}{\partial E}E\frac{\partial\Theta}{\partial E}, \\
\frac{\partial f}{\partial \hat{p}^i} &= -E\frac{\partial\bar{f}}{\partial E}\frac{\partial\Theta}{\partial\hat{p}^i}, \\
\frac{\partial f}{\partial x^i} &= -E\frac{\partial\bar{f}}{\partial E}\frac{\partial\Theta}{\partial x^i}.
\end{aligned} \tag{6.2}$$

Plugging (1) into the LHS of Boltzmann equation (6.1) at zero order we get the redshift $T\propto a^{-1}$ of photon temperature

$$\cancel{\tfrac{E}{a}E\partial_E\bar{f}}\left[\tfrac{1}{\bar{T}}\tfrac{d\bar{T}}{d\tau} + \mathcal{H}\right] = 0 \quad\Rightarrow\quad \bar{T}\propto\frac{1}{a}.$$

At first order, replacing $\frac{\dot{T}}{\bar{T}} = -\mathcal{H}$, we get

$$-\tfrac{E}{a}E\partial_E\bar{f}\left[\frac{\partial\Theta}{\partial\tau} + \hat{p}^j\frac{\partial\Theta}{\partial x^j} - \mathcal{H}E\cancel{\frac{\partial\Theta}{\partial E}} + \hat{p}^j\frac{\partial\psi}{\partial x^j} - \frac{\partial\phi}{\partial\tau}\right] = C[f]. \tag{6.3}$$

At late times collisions with baryons are essentially low-energy Compton scatterings. To good approximation, these scatterings do not change the energy of photons but only the direction of their momentum, and we can thus assume $\Theta = \Theta(\tau, x^i, \hat{p}^i)$, i.e. the temperature perturbation is independent of the photon energy.

1.1 Collision term from Compton scattering

We need to compute the collision term $C[f]$ for Compton scatterings

1.2 Passing to multipoles

We can now pass to Fourier space and decompose the angular dependence of Θ in multipoles to get a hierarchy of coupled ODEs for the multipole moments Θ_ℓ. Passing to Fourier space in (6.3), we get

Fix RHS

$$\left[\frac{\partial\Theta}{\partial\tau} + i\vec{k}\cdot\hat{p}\,\Theta - \mathcal{H}E\cancel{\frac{\partial\Theta}{\partial E}} + i\hat{p}\cdot\vec{k}\,\psi - \frac{\partial\phi}{\partial\tau}\right] = -\frac{1}{\frac{E}{a}E\partial_E\bar{f}}C[f].$$

The dependence on $\vec{k}$ only enter via the modulus k and the angle $\mu := \hat{k}\cdot\hat{p}$, so we can decompose the angular dependence in Legendre polynomials $P_\ell(\mu)$ as

$$\Theta(\tau,k,\mu) = \sum_{\ell=0}^{\infty}\frac{2\ell+1}{i^\ell}\Theta_\ell(\tau,k)P_\ell(\mu)\,, \quad \Theta_\ell(\tau,k) = \frac{i^\ell}{2}\int_{-1}^{1}d\mu\,P_\ell(\mu)\Theta(\tau,k,\mu).$$

The first three multipoles have a simple physical interpretation: up to numerical factors, Θ_0 is the monopole i.e. the average temperature perturbation; Θ_1 is the dipole i.e. the bulk velocity of the photon fluid; Θ_2 is the quadrupole i.e. the anisotropic stress of the photon fluid. Higher multipoles Θ_ℓ with $\ell \geqslant 3$ encode higher-order angular correlations of the temperature perturbation, and are generated by free-streaming of photons after decoupling.

Recall Legendre multipoles are orthogonal, satisfy a recursion relation and the first 3 are

$$\int_{-1}^{1}d\mu\,P_\ell(\mu)P_{\ell'}(\mu) = \tfrac{2}{2\ell+1}\delta_{\ell\ell'},\ \mu P_\ell(\mu) = \tfrac{\ell+1}{2\ell+1}P_{\ell+1}(\mu) + \tfrac{\ell}{2\ell+1}P_{\ell-1}(\mu),\ P_0(\mu) = 1,\ P_1(\mu) = \mu,\ P_2(\mu) = \frac{3\mu^2-1}{2}.$$

Multiplying the equation by $\frac{i^\ell}{2}P_\ell(\mu)$ and integrating over μ, using the orthogonality and recursion relations and the fact that ψ and ϕ are independent of μ, we get

$$\frac{\partial\Theta_\ell}{\partial\tau} + k\left(\frac{\ell+1}{2\ell+1}\Theta_{\ell+1} - \frac{\ell}{2\ell+1}\Theta_{\ell-1}\right) - \mathcal{H}E\cancel{\frac{\partial\Theta_\ell}{\partial E}} - \tfrac{1}{3}k\psi\delta_{\ell1} - \frac{\partial\phi}{\partial\tau}\delta_{\ell0} = -\frac{1}{\frac{E}{a}E\partial_E\bar{f}}C_\ell[f],$$

Fix RHS

The Boltzmann equation then becomes a hierarchy of coupled ODEs for multipole moments Θ_ℓ, which can be solved numerically to get the CMB power spectrum and other observables,

$$\begin{aligned}
\frac{\partial\Theta_0}{\partial\log a} - E\cancel{\frac{\partial\Theta_0}{\partial E}} - \frac{\partial\phi}{\partial\log a} + \frac{k}{\mathcal{H}}\Theta_1 &= -\frac{1}{\mathcal{H}}\frac{1}{\frac{E}{a}E\partial_E\bar{f}}C_0[f],\\
\frac{\partial\Theta_1}{\partial\log a} - E\cancel{\frac{\partial\Theta_1}{\partial E}} + \tfrac{2}{3}\frac{k}{\mathcal{H}}\Theta_2 - \tfrac{1}{3}\frac{k}{\mathcal{H}}\Theta_0 - \tfrac{1}{3}\frac{k}{\mathcal{H}}\psi &= -\frac{1}{\mathcal{H}}\frac{1}{\frac{E}{a}E\partial_E\bar{f}}C_1[f],\\
\frac{\partial\Theta_2}{\partial\log a} - E\cancel{\frac{\partial\Theta_2}{\partial E}} + \tfrac{3}{5}\frac{k}{\mathcal{H}}\Theta_3 - \tfrac{2}{5}\frac{k}{\mathcal{H}}\Theta_1 &= -\frac{1}{\mathcal{H}}\frac{1}{\frac{E}{a}E\partial_E\bar{f}}C_2[f],\\
\frac{\partial\Theta_\ell}{\partial\log a} - E\cancel{\frac{\partial\Theta_\ell}{\partial E}} + \frac{k}{\mathcal{H}}\left(\tfrac{\ell+1}{2\ell+1}\Theta_{\ell+1} - \tfrac{\ell}{2\ell+1}\Theta_{\ell-1}\right) &= -\frac{1}{\mathcal{H}}\frac{1}{\frac{E}{a}E\partial_E\bar{f}}C_\ell[f], \quad \ell \geqslant 3.
\end{aligned}$$

We passed to time variable $\log a$ to highlight dominant terms in the super(sub)-ubhorizon regime $k/\mathcal{H} \ll 1$. Note only the first 3 equations for $\Theta_0, \Theta_1, \Theta_2$ are sourced. Metric potential enter only the first 2, while anisotropic stress enters the last one.

The hierarchy is infinite and thus needs to be truncated at some ℓ_{max}. A convenient way is to impose some boundary condition at ℓ_{max}, for example $\Theta_{\ell_{\text{max}}+1} = \Theta_{\ell_{\text{max}}}$ or $\Theta_{\ell_{\text{max}}+1} = 0$, which is a good approximation since higher multipoles are generated by free-streaming and thus are suppressed by powers of $k/\mathcal{H}$ at early times when the hierarchy is sourced

write eq from Whinter

...

Recalling that $\Theta_0 = \frac{\delta T_\gamma}{T_\gamma} = \frac{1}{4}\delta_\gamma$ as confirmed by $\rho_\gamma \propto T_\gamma^4$ and $\Theta_1 = -\frac{1}{3}v_\gamma$, we see the first 2 equations are simply the continuity and Euler equations for photons, while the last one is the evolution equation for photon anisotropic stress

...

2 The photon-baryon fluid

The relevant interactions for the photon-baryon fluid are

$$\begin{aligned}
e^- + p \leftrightarrow e^- + p, &\quad \text{(Coulomb scattering)}\\
e^- + \gamma \leftrightarrow e^- + \gamma, &\quad \text{(Thomson scattering:=low-energy Compton scattering)}\\
p + \gamma \leftrightarrow p + \gamma, &\quad \text{(Thomson scattering:=low-energy Compton scattering)}
\end{aligned}$$

In principle we would have 3 separate Boltzmann equations for electrons, protons and photons with corresponding collision terms

$$\begin{aligned}
\frac{df_e}{d\lambda} &= C_{e\gamma}[f_e, f_\gamma] + C_{ep}[f_e, f_p],\\
\frac{df_p}{d\lambda} &= C_{p\gamma}[f_p, f_\gamma] + C_{pe}[f_p, f_e],\\
\frac{df_\gamma}{d\lambda} &= C_{\gamma e}[f_\gamma, f_e] + C_{\gamma p}[f_\gamma, f_p].
\end{aligned}$$

Coulomb scattering is extremely efficient at keeping electrons and protons tightly coupled at *all times*, allowing to treat them as a single baryon fluid with common bulk quantities to extreme accuracy

$$\delta_e = \delta_p = \delta_b, \quad \vec{u}_e = \vec{u}_p = \vec{u}_b.$$

The second interaction keeps photons tightly coupled to electrons, and thus to protons, making sense of the photon-baryon fluid picture at early times. The last interaction is negligible since the Thompson cross-section $\sigma_T \propto m^{-2}$ and thus for protons is suppressed by a factor $(m_e/m_p)^2$.

Thompson scattering does not change number density, but does exchange momentum between photons and baryons, and thus appears as a source term in the Euler equations of the two fluids, with opposite signs to ensure total momentum conservation. That is, the conservation equations read

$$\nabla_\mu T^\mu_{(\gamma)\,0} = \nabla_\mu T^\mu_{(b)\,0} = 0 \quad \text{while} \quad \nabla_\mu T^\mu_{(\gamma)\,j} = -\nabla_\mu T^\mu_{(b)\,j} =: Q_j \neq 0.$$

Regardless of the interaction, the RHS for the Euler equations is written in terms of the velocity $v_j = \frac{1}{(\rho+p)}T^0_j$. Therefore if we know the RHS of either the photon or baryon Euler equation, we can immediately get the RHS of the other by

$$\frac{\partial v^j_\gamma}{\partial \tau} + \ldots = \frac{1}{(\rho_\gamma + p_\gamma)} Q_j =: X \quad \Rightarrow \quad \frac{\partial v^j_b}{\partial \tau} + \ldots = -\frac{1}{(\rho_b + p_b)} Q_j = -\frac{(\rho_\gamma + p_\gamma)}{(\rho_b + p_b)} \frac{1}{(\rho_\gamma + p_\gamma)} Q_j = -\frac{4}{3}\frac{\rho_\gamma}{\rho_b} X.$$

3 CMB Lensing

4 CMB Power spectrum

We expand $\Theta(\vec{x}, \hat{p}, t)$ in spherical harmonics and Fourier components

$$a_{\ell m}(\vec{x}, t) = \int d\Omega_{\hat{p}}\, Y^*_{\ell m}(\hat{p}) \Theta(\vec{x}, \hat{p}, t)) = \int \frac{d^3k}{(2\pi)^3} e^{i\vec{k}\cdot\vec{x}} \int d\Omega_{\hat{p}} Y^*_{\ell m}(\hat{p}) \Theta(\vec{k}, \hat{p}, t).$$

Inflation predicts that the initial perturbations (like $\mathcal{R}$) form a random field that is very close to Gaussian. Consequently Θ and the $a_{\ell m}$ will also form a Gaussian random field with mean zero and some variance C_ℓ, i.e.

$$\langle a_{\ell m} a^*_{\ell' m'} \rangle = \delta_{\ell\ell'} \delta_{mm'} C_\ell$$

where the brackets denote an ensemble average (average over many different realizations of our Universes). And since it is Gaussian all the statistical information about Θ is contained in C_ℓ (in practice there will be some deviations from pure Gaussianity depending on the model of inflation so higher order moments do potentially contain interesting information). For a fixed ℓ, each $a_{\ell m}$ has the same variance and for each value of ℓ there are $2\ell + 1$ possible values of m. Thus in an experiment after measuring the $a_{\ell m}$'s we can estimate the power-spectrum as

$$\hat{C}_\ell = \frac{1}{2\ell + 1} \sum_{m=-\ell}^{\ell} |a_{\ell m}|^2$$

There is an important difference with what we measure and what the theory says. The theory is formulated in terms of an ensemble average (average of many Universes) whereas we only have one Universe to measure in. This is where the ergodic assumption comes into play, which tells us that as long as we have enough modes the mean of these will be the same as the ensemble average. This means that for large ℓ's we have enough statistical power to accurately estimate C_ℓ, while for very low ℓ we will run into cosmic variance: we have very few m's to estimate C_ℓ from so there is a fundamental uncertainty that is due to us only having one single Universe to measure this in that we cannot get around. Since $\hat{C}_\ell$ is the sum of squared Gaussians it follows a χ^2 distribution with uncertainty given by

$$\frac{\mathrm{Var}(\hat{C}_\ell)}{C_\ell^2} = \frac{2}{2\ell + 1}, \quad \text{with} \quad \hat{C}_\ell = \frac{1}{2\ell + 1} \sum_m |a_{\ell m}|^2 \sim \ \chi^2 \text{ distributed}$$

This uncertainty is what is depicted by the green shaded region (and the large red error-bars) for low ℓ in Figure 1 and shows that our current measurements for temperature fluctuations are close to as good as we can get.

To derive an expression for C_ℓ we expand Θ in multipoles

$$a_{\ell m}(\vec{x}, t) = \sum_{\ell_1} (2\ell_1 + 1)(-i)^{\ell_1} \int \frac{d^3k}{(2\pi)^3} e^{i\vec{k}\cdot\vec{x}} \int d\Omega_{\hat{p}}\, Y^*_{\ell m}(\hat{p}) P_{\ell_1}(\mu) \Theta_{\ell_1}(\vec{k}, t)$$

where $\mu = \hat{p} \cdot \hat{k}$ which gives us

$$a_{\ell m} a^*_{\ell' m'} = \sum_{\ell_1, \ell_2} (2\ell_1 + 1)(2\ell_2 + 1)(-i)^{\ell_1 - \ell_2} \int \frac{d^3k\, d^3k'}{(2\pi)^6} e^{i(\vec{k} - \vec{k}')\cdot\vec{x}}$$
$$\int d\Omega_{\hat{p}} d\Omega_{\hat{p}'} Y^*_{\ell m}(\hat{p}) Y_{\ell' m'}(\hat{p}') P_{\ell_1}(\mu) P_{\ell_2}(\mu') \Theta_{\ell_1}(\vec{k}, t) \Theta^*_{\ell_2}(\vec{k}', t).$$

Now remember that when we solve this numerically we put $\Theta_\ell(\vec{k}, t) = \Theta^{\mathrm{code}}_\ell(k, t) \mathcal{R}_{\mathrm{ini}}(\vec{k})$ where the initial value of the curvature perturbation $\mathcal{R}_{\mathrm{ini}} \propto \Psi$ was set to unity, i.e. we factored out the common initial condition from inflation before solving (which is perfectly fine since the equation system is linear). We must now put it back in. When we take the ensemble average of the above this will lead to a term

$$\langle \mathcal{R}_{\mathrm{ini}}(\vec{k}) \mathcal{R}^*_{\mathrm{ini}}(\vec{k}') \rangle = (2\pi)^3 \delta(\vec{k} - \vec{k}') P(k)$$

where the primordial power-spectrum is given by

$$P(k) = \frac{2\pi^2}{k^3}\mathcal{P}_{\mathcal{R}}(k), \qquad \mathcal{P}_{\mathcal{R}}(k) = A_s(k/k_{\rm pivot})^{n_s-1}\,.$$

The delta-function will enforce $\vec{k} = \vec{k}'$ taking care of one of the integrals and

$$\langle a_{\ell m} a^*_{\ell' m'}\rangle = \sum_{\ell_1,\ell_2}(2\ell_1+1)(2\ell_2+1)(-i)^{\ell_1-\ell_2}\int\frac{d^3k}{(2\pi)^3}\int d\Omega_{\hat{p}}d\Omega_{\hat{p}'}$$
$$\cdot\, Y^*_{\ell m}(\hat{p})Y_{\ell' m'}(\hat{p}')P_{\ell_1}(\mu)P_{\ell_2}(\mu')\Theta^{\rm code}_{\ell_1}(k,t)\Theta^{\rm code*}_{\ell_2}(k,t)\frac{2\pi^2\mathcal{P}_{\mathcal{R}}(k)}{k^3}$$

For the next step we need

$$\int d\Omega_{\hat{p}}\, P_\ell(\mu)Y^*_{\ell' m}(\hat{p}) = \delta_{\ell\ell'}\frac{4\pi}{2\ell+1}Y^*_{\ell m}(\hat{k})$$

which takes care of two of the integrals (and the two Kronecker deltas kill both of the sums) leaving us with

$$\langle a_{\ell m} a^*_{\ell' m'}\rangle = \delta_{\ell\ell'}(4\pi)^2\int\frac{d^3k}{(2\pi)^3}\,Y_{\ell m}(\hat{k})Y^*_{\ell m'}(\hat{k})|\Theta^{\rm code}_\ell(k,t)|^2\frac{2\pi^2\mathcal{P}_{\mathcal{R}}(k)}{k^3}$$

Finally splitting the k-integral into an integral over angles ($\Omega_{\hat{k}}$ now means over the angles of k) using that

$$\int d\Omega_{\hat{k}}\, Y_{\ell m}(\hat{k})Y^*_{\ell m'}(\hat{k}) = \delta_{mm'}$$

we get

$$\langle a_{\ell m} a^*_{\ell' m'}\rangle = \delta_{\ell\ell'}\delta_{mm'}4\pi\int\frac{dk}{k}\,\mathcal{P}_{\mathcal{R}}(k)|\Theta^{\rm code}_\ell(k,t)|^2$$

so

$$C_\ell = 4\pi\int\mathcal{P}_{\mathcal{R}}(k)|\Theta^{\rm code}_\ell(k)|^2\frac{dk}{k}$$

where the time is taken to be today (which is the time when we observe the CMB, so $\Theta^{\rm code}_\ell(k) = \Theta^{\rm code}_\ell(k, t = t_{\rm today})$). The difference with respect to what is in Chapter 8.5.2 in Dodelson is that he expresses this in terms of

$$P(k) = \frac{2\pi^2}{k^3}\mathcal{P}_{\mathcal{R}}(k)$$

which gives us the equivalent formulation

$$C_\ell = \frac{2}{\pi}\int k^2P(k)|\Theta^{\rm code}_\ell(k)|^2dk$$

The expression above tells us how to go from the stuff we computed numerically, the temperature multipoles Θ_ℓ, to the theoretical prediction for the CMB angular power-spectrum. This can be directly compared to the equivalent quantity measured in experiments.

7 Neutrinos in Cosmology

The treatment of neutrios is almost identical to that of photons, since they can be treated as a relativistic fluid down to very late times. We still assume they retain local thermal equilibrium with a Fermi-Dirac distribution with temperature constrast $\mathcal{N}$, that is

$$f_\nu(\tau, \vec{x}, \vec{p}) = \frac{1}{e^{(E-\mu)/T_\nu} + 1}, \quad T_\nu = \bar{T}_\nu(\tau)(1 + \mathcal{N}(\tau, \vec{x}, \hat{p}, E)).$$

The main differences are the following.

- Neutrino temperature is lower than the photon temperature $\bar{T}_\nu \simeq (4/11)^{1/3}\bar{T}_\gamma$, since neutrinos decouple before electron-positron annihilation, which heats up photons but not neutrinos. The treatment is however unchanged, since the Boltzmann equation only depends on the temperature contrast $\delta T/T$, not on the absolute value.

- Neutrinos are fermions, thus following Fermi-Dirac distribution, while photons are bosons, thus with Bose-Einstein distribution. Again this difference is neglible.

- Neutrino have no interactions after decoupling, and the RHS of the Boltzmann equation is thus zero. In particular they free-stream and develop non-negligible anisotropic stress (Legendre multipole $\mathcal{N}_2$ and possibly higher ones) already at early times, while photons are driven towards the first two multipoles Θ_0, Θ_1 by Thomson scattering until recombination and develop anisotropic stress only after decoupling. Neutrino anisotropic stress has a non-negligible effect on the evolution of metric perturbations, and thus on the CMB power spectrum and other observables.

- Neutrino are actually massive, and become non-relativistic at late times, while photons are always relativistic. This affects the background evolution of the universe, and thus the expansion history and the growth of structure at late times.

1 Geodesic & Boltzmann Equations for neutrinos

The geodesic equations in newtonian gauge are unchanged

$$\begin{aligned}
\frac{dE}{d\tau} &= -\frac{p^2}{E}(\mathcal{H} - \dot{\phi}) - p\hat{p}^j\partial_j\psi, \\
\frac{dp}{d\tau} &= -p(\mathcal{H} - \dot{\phi}) - E\hat{p}^j\partial_j\psi, \\
\frac{d\hat{p}^i}{d\tau} &= \left(\hat{p}^i\hat{p}^j - \delta^{ij}\right)\left(\frac{E}{p}\partial_j\psi + \frac{p}{E}\partial_j\phi\right).
\end{aligned} \tag{7.1}$$

Beware taking $E = p$ is a perfect approximation down to times/temperatures $T_\nu \sim 1\,\text{eV} \simeq 1.6 \times 10^4$ K, but will eventually break down at later times. Also recall neutrinos' temperature is related to photons by $T_\nu = \left(\frac{4}{11}\right)^{1/3}T_\gamma$. Ineed from neutrino oscillations we know that at leaste one neutrino mass $m_\nu \gtrsim 0.05$ eV, while from cosmological bounds we know that the sum of neutrino masses $\sum m_\nu \lesssim 0.1$eV.

The last equation implies that to zeroth order in metric perturbations, neutrinos travel along straight lines, while at first order they are deflected by the gravitational potentials ψ and ϕ. The first two equations encode the change in energy and momentum along the trajectory.

Conidering the relativistic limit $E = p$ for simplicity it is convenient to trade partial with total time derivatives $\frac{\partial\phi}{\partial\tau} = \frac{d\phi}{d\tau} - \frac{\partial\phi}{\partial x^j}\frac{dx^j}{d\tau}$ to get,

$$\frac{d\log E}{d\tau} = -\mathcal{H} + \frac{d\phi}{d\tau} - \hat{p}^j(\partial_j\phi + \partial_j\psi).$$

To order zero it encode the expected redshift behavior $E = p = \propto a^{-1}$. The perturbed part implies energy is red-blueshifted by the gravitational potentials via the istantaneous and integrated Sachs-Wolf effect. Since the change in $\hat{p}^j$ along the trajectory is already first order, we can integrate it to get

$$\begin{aligned}\log\left(\frac{E_f}{E_i}\right) &= \int_{\tau_i}^{\tau_f} d\tau \; -\mathcal{H} + \frac{d\phi}{d\tau} - \hat{p}^j(\partial_j\phi + \partial_j\psi)\, d\tau \\ &= \log\left(\frac{a_i}{a_f}\right) + \underbrace{\big(\phi(\tau_f, x_f) - \phi(\tau_i, x_i)\big)}_{\text{instantaneous SW (aka gravitational redshift)}} - \int_{\tau_i}^{\tau_f} \underbrace{d\tau\; \hat{p}^j(\partial_j\phi + \partial_j\psi)}_{\text{integrated Sachs-Wolfe effect}} .\end{aligned}$$

Writing $E = \bar{E}(1+\mathcal{N})$ with $\bar{E}\propto a^{-1}$ so that $\bar{E}_i a_i = \bar{E}_f a_f$ to remove the background redshift, we get the expected Sachs-Wolfe formula for the observed temperature anisotropies

$$\mathcal{N}_f(\hat{p}) - \mathcal{N}_i(\hat{p}) \simeq \log\left(\frac{1+\mathcal{N}_f(\hat{p})}{1+\mathcal{N}_i(\hat{p})}\right) = \phi(\tau_f, x_f) - \phi(\tau_i, x_i) - \int_{\tau_i}^{\tau_f} d\tau\; \hat{p}^j(\partial_j\phi + \partial_j\psi).$$

The above formulas are exact in the relativistic limit, but nedd to be adjusted using the geodesic equations (7.1) with $E \neq p$ to get the effect of neutrino masses at late times.

The Boltzmann equation (4.9) for neutrinoes, accounting for masses, is again (second order in metric perturbations, exact otherwise)

$$\begin{aligned}C[f] = \frac{df}{d\lambda} &= P^0\left[\frac{\partial f}{\partial\tau} + \frac{\partial f}{\partial x^i}\frac{P^i}{P^0} + \frac{\partial f}{\partial E}\frac{1}{P^0}\frac{dE}{d\lambda} + \frac{\partial f}{\partial\hat{p}^i}\frac{1}{P^0}\frac{d\hat{p}^i}{d\lambda}\right] \\ &= \frac{E}{a}\left[\frac{\partial f}{\partial\tau}(1-\psi) + \left(\hat{p}^j\partial_j f\right)\frac{p}{E}(1+\phi) - \frac{\partial f}{\partial E}\frac{p^2}{E}\left(\mathcal{H}(1-\psi) - \dot{\phi}\right) - \frac{\partial f}{\partial\bar{p}^j}\partial_j\psi\, E + \left(\frac{\partial f}{\partial p}\hat{p}^j - \frac{\partial f}{\partial\bar{p}^j}\right)\partial_j\phi\,\frac{p^2}{E}\right].\end{aligned}$$

Discuss

discuss

Neutrinos retain local equilibrium with a Fermi-Dirac distribution down to late times, simply with a time and momentum dependent temperature, and the chemical potential is negligible

$$f(\tau, x^i, E, \hat{p}^i) = \frac{1}{\exp\left(\frac{E}{T_\nu}\right) + 1}, \quad T_\nu = \bar{T}_\nu(\tau)(1 + \mathcal{N}(\tau, x^i, p, \hat{p}^i)) = \bar{T}_\nu + \delta T_\nu, \quad \mu \sim 0.$$

More generally, assuming the full distribution is still a function $f(E/T)$ of the ratio E/T to some *local* temperature T, we Taylor expand the distribution function and its derivative just like for photons and rewrite the Boltzmann equation. At zero order we get the redshift $T\propto a^{-1}$ of neutrino temperature

$$\cancel{\tfrac{E}{a}E\partial_E\bar{f}}\left[\tfrac{1}{\bar{T}}\tfrac{d\bar{T}}{d\tau} + \mathcal{H}\right] = 0 \quad \Rightarrow \quad \bar{T}_\nu\propto\frac{1}{a}.$$

At first order, replacing $\frac{\dot{T}}{\bar{T}} = -\mathcal{H}$, we get

$$-\tfrac{E}{a}E\partial_E\bar{f}\left[\frac{\partial\mathcal{N}}{\partial\tau} + \hat{p}^j\frac{\partial\mathcal{N}}{\partial x^j} - \mathcal{H}E\frac{\partial\mathcal{N}}{\partial E} + \hat{p}^j\frac{\partial\psi}{\partial x^j} - \frac{\partial\phi}{\partial\tau}\right] = 0. \tag{7.2}$$

Note there is indeed no right-hand side, since neutrinos have no sizable interactions after decoupling. On the other hand, we cannot drop the term $\mathcal{H}E\frac{\partial\mathcal{N}}{\partial E}$ as we did for photons, since neutrinos masses will eventually make $\mathcal{N}$ depend on E at late times.

1.1 Passing to multipoles

We now pass to Fourier space and decompose the angular dependence of $\mathcal{N}$ in multipoles to get a hierarchy of coupled ODEs for the multipole moments $\mathcal{N}_\ell$. Passing to Fourier space in (7.2), we get

$$\left[\frac{\partial\mathcal{N}}{\partial\tau} + i\vec{k}\cdot\hat{p}\mathcal{N} - \mathcal{H}E\frac{\partial\mathcal{N}}{\partial E} + i\hat{p}\cdot\vec{k}\,\psi - \frac{\partial\phi}{\partial\tau}\right] = 0.$$

The dependence on $\vec{k}$ only enter via the modulus k and the angle $\mu := \hat{k}\cdot\hat{p}$, so we can decompose the angular dependence in Legendre polynomials $P_\ell(\mu)$ as

$$\mathcal{N}(\tau, k, \mu) = \sum_{\ell=0}^{\infty}\frac{2\ell+1}{i^\ell}\mathcal{N}_\ell(\tau, k)P_\ell(\mu)\,, \quad \mathcal{N}_\ell(\tau, k) = \frac{i^\ell}{2}\int_{-1}^{1} d\mu\, P_\ell(\mu)\mathcal{N}(\tau, k, \mu).$$

The first three multipoles have a simple physical interpretation: up to numerical factors, $\mathcal{N}_0$ is the monopole i.e. the average temperature perturbation; $\mathcal{N}_1$ is the dipole i.e. the bulk velocity of the neutrino fluid; $\mathcal{N}_2$ is the quadrupole i.e. the anisotropic stress of the neutrino fluid. Higher multipoles $\mathcal{N}_\ell$ with $\ell \geqslant 3$ encode higher-order angular correlations of the temperature perturbation, and are generated by free-streaming of neutrinos after decoupling.

Recall Legendre multipoles are orthogonal, satisfy a recursion relation and the first 3 are

$$\int_{-1}^{1} d\mu\, P_\ell(\mu) P_{\ell'}(\mu) = \tfrac{2}{2\ell+1}\delta_{\ell\ell'},\ \ \mu P_\ell(\mu) = \tfrac{\ell+1}{2\ell+1}P_{\ell+1}(\mu) + \tfrac{\ell}{2\ell+1}P_{\ell-1}(\mu),\ \ P_0(\mu) = 1,\ \ P_1(\mu) = \mu,\ \ P_2(\mu) = \frac{3\mu^2-1}{2}.$$

Multiplying the equation by $\frac{i^\ell}{2}P_\ell(\mu)$ and integrating over μ, using the orthogonality and recursion relations, and the fact that ψ and ϕ are independent of μ, we get

$$\frac{\partial \mathcal{N}_\ell}{\partial \tau} + k\left(\frac{\ell+1}{2\ell+1}\mathcal{N}_{\ell+1} - \frac{\ell}{2\ell+1}\mathcal{N}_{\ell-1}\right) - \mathcal{H}E\frac{\partial \mathcal{N}_\ell}{\partial E} - \tfrac{1}{3}k\psi\delta_{\ell 1} - \frac{\partial \phi}{\partial \tau}\delta_{\ell 0} = 0.$$

The Boltzmann equation becomes a hierarchy of coupled differential equations for multipole moments $\mathcal{N}_\ell$, which can be solved numerically to get the evolution of neutrino perturbations. Notably, because of masses we retain the energy terms $E\frac{\partial \mathcal{N}_\ell}{\partial E}$ and we get linear first-order PDEs instead of ODEs as for photons. We could then Fourier transform the energy dependence and solve the resulting system of ODEs for each value of the Fourier variable, albeit computationally expensive. Another approach is to average over the energy dependence to get a system of ODEs for the energy-averaged multipoles, which is computationally cheaper but less accurate at late times (cf. [MB95] for details).

The hierarchy is anyway infinite and thus needs to be truncated at some ℓ_{max} just like for photons, albeit with a higher ℓ_{max} since neutrinos free-stream and develop higher multipoles already at early times, while photons are driven towards the first two multipoles by Thomson scattering until recombination and develop higher multipoles only after decoupling. A convenient way is to impose some boundary condition at ℓ_{max}, for example $\mathcal{N}_{\ell_{\text{max}}+1} = \mathcal{N}_{\ell_{\text{max}}}$ or $\mathcal{N}_{\ell_{\text{max}}+1} = 0$, which is a good approximation since higher multipoles are generated by free-streaming and thus are suppressed by powers of $k/\mathcal{H}$ at early times when the hierarchy is sourced

Shorten & refer to γ section

write eq from Whinter

...

Passing to variable $\log a$ to highlight dominant terms in the super(sub)-ubhorizon regime $k/\mathcal{H} \ll 1$ we get

$$\begin{aligned}
\frac{\partial \mathcal{N}_0}{\partial \log a} - E\frac{\partial \mathcal{N}_0}{\partial E} - \frac{\partial \phi}{\partial \log a} + \frac{k}{\mathcal{H}}\mathcal{N}_1 &= 0,\\
\frac{\partial \mathcal{N}_1}{\partial \log a} - E\frac{\partial \mathcal{N}_1}{\partial E} + \tfrac{2}{3}\frac{k}{\mathcal{H}}\mathcal{N}_2 - \tfrac{1}{3}\frac{k}{\mathcal{H}}\mathcal{N}_0 - \tfrac{1}{3}\frac{k}{\mathcal{H}}\psi &= 0,\\
\frac{\partial \mathcal{N}_2}{\partial \log a} - E\frac{\partial \mathcal{N}_2}{\partial E} + \tfrac{3}{5}\frac{k}{\mathcal{H}}\mathcal{N}_3 - \tfrac{2}{5}\frac{k}{\mathcal{H}}\mathcal{N}_1 &= 0,\\
\frac{\partial \mathcal{N}_\ell}{\partial \log a} - E\frac{\partial \mathcal{N}_\ell}{\partial E} + \frac{k}{\mathcal{H}}\left(\tfrac{\ell+1}{2\ell+1}\mathcal{N}_{\ell+1} - \tfrac{\ell}{2\ell+1}\mathcal{N}_{\ell-1}\right) &= 0, \quad \ell \geqslant 3.
\end{aligned}$$

Note only the first 2 equations for $\mathcal{N}_0, \mathcal{N}_1$ are sourced (by metric potentials), while the 3rd one describe the evolution of the neutrino anisotropic stress $\mathcal{N}_2$ which is nonnegligible (for accuracy below 10%) on small scales even at early times.

Recalling that $\mathcal{N}_0 = \frac{\delta T_\nu}{T_\nu} = \frac{1}{4}\delta_\nu$ as confirmed by $\rho_\nu \propto T_\nu^4$ and $\mathcal{N}_1 = -\frac{1}{3}v_\nu$, we see the first 2 equations are simply the continuity and Euler equations for neutrinos, while the last one is the evolution equation for neutrino anisotropic stress

Check & finish

...

2 Neutrino masses

2.1 Other bounds

2.1.1 Lower bound from neutrino oscillations.

Neutrino oscillation experiments determine *mass-squared differences* $\Delta m_{21}^2 \equiv m_2^2 - m_1^2$ and $\Delta m_{31}^2 \equiv m_3^2 - m_1^2$ (and the mixing angles), but they do *not* fix the absolute mass scale. (i.e. they are insensitive to adding a common

mass to all eigenstates). Nevertheless, since the measured splittings are nonzero, at least one neutrino mass eigenvalue must satisfy

$$m_{\text{heavy}} \gtrsim \sqrt{|\Delta m_{31}^2|} \sim 0.05\text{eV},$$

so neutrinos cannot all be massless. Using the oscillation data and taking the lightest mass to zero gives a robust *minimum* for the sum of masses:

$$\Sigma m_\nu \equiv m_1 + m_2 + m_3 \gtrsim \begin{cases} 0.058 \text{ eV}, & \text{normal ordering (NO)}, \\ 0.098 \text{ eV}, & \text{inverted ordering (IO)}. \end{cases}$$

2.2 Bounds from cosmology

We can get bound on neutrino masses from cosmology quite easily. For example, since neutrino eventually become nonrelativistic matter, they contribute to the matter density today and we must have

$$\Omega_\nu = \frac{\rho_\nu}{\rho_{c0}} \simeq \frac{m_\nu n_\nu}{\rho_{c0}} \leqslant \Omega_m \simeq 0.3$$

The approximation $\rho_\nu \simeq m_\nu n_\nu$ is valid at late times $T_\nu \ll m_\nu$ and can be made more precise integrating the Fermi-Dirac distribution and expanding in T_ν/m_ν to get

$$\rho_\nu = m_\nu n_\nu \left(1 + \frac{3}{2}\frac{T_\nu}{m_\nu} + O\Big(\frac{T_\nu^2}{m_\nu^2}\Big)\right).$$

Since neutrinos decouple around $T_\nu = T_\gamma \simeq 1\,\text{MeV}$, their number density just dilute as $n_\nu \propto a^{-3}$ afterward, and we later get $T_\nu = (4/11)^{1/3} T_\gamma$ due to electron-positron annihilation heating up photons but not neutrinos, so that[1]

$$n_\nu(a_0) = \tfrac{3}{4}\tfrac{3}{2}\Big(\frac{T_\nu}{T_\gamma}\Big)^3 n_\gamma(a_0) = \tfrac{3}{4}\tfrac{3}{2}\tfrac{3}{11} n_\gamma(a_0) = \frac{27}{88}\frac{2\zeta(3)}{\pi^2} T_{\gamma 0}^3.$$

Up to numerical factors of $O(1)$, after plugging in numbers, we get

$$m_\nu \lesssim \frac{\Omega_m \rho_{c0}}{n_{\nu 0}} \sim \frac{\Omega_{m0}\frac{3H_0^2}{8\pi G}}{\frac{1}{10}T_{\gamma 0}^3} \quad \Rightarrow \quad m_\nu \lesssim 0.5\,\text{eV}.$$

This is just a rough estimate, but more careful analyses using CMB and LSS data give a bound of $\sum m_\nu \lesssim 0.1$ eV on the sum of neutrino masses, which is already quite close to the lower bound from oscillation experiments.

[1] The factor $\frac{3}{4}$ is from Fermi vs Bose statistics of neutrinos and photons, while the factor $\frac{3}{2}$ is from 3 neutrino flavors vs 2 photon polarizations.

8 Inflation & Initial Conditions

1 Problems solved by inflations

Inflation is a period of accelerated expansion in the early universe, characterized by nearly exponential growth of the scale factor with Hubble parameter H approximately constant,

$$H = \frac{\frac{da}{dt}}{a} \approx \text{const}, \quad \mathcal{H} = aH = He^{Ht} = -\tfrac{1}{\tau}, \quad a(t) \simeq e^{H_* t} = -\tfrac{1}{H\tau} \quad \text{with} \quad t \in \mathbb{R},\ \tau \in (-\infty, 0).$$

Conformal time τ is negative during inflation approaching zero as $t \to \infty$. Notably the comoving Hubble radius $d_{\mathcal{H}} := \frac{1}{\mathcal{H}} = a^{-1}\frac{1}{H}$ shrinks over time. Beware the comoving particle horizon d_{hor} grows even during inflation, albeit approaching a constant value as $a \to \infty$, while the physical particle horizon grows to infinity in any case as physically expected,

$$d_{\text{hor}}(a) = \int_{\tau_{in}}^{\tau} c\, d\tau = \int_{a_{in}}^{a} \frac{da}{a^2 H} = \frac{1}{H}\left(\frac{1}{a_{in}} - \frac{1}{a}\right) \to \frac{1}{H}\frac{1}{a_{in}}, \quad d_{\text{hor}}^{\text{phys}} = a\, d_{\text{hor}} = \frac{1}{H}\left(\frac{a}{a_{in}} - 1\right) \to \infty \quad \text{as} \quad a \to \infty.$$

Inflation solves several problems of the standard Big Bang cosmology.

- The exponential expansion drives the universe towards flatness, thus solving the flatness problem.
- Exponential expansion also dilutes any unwanted relics, such as magnetic monopoles, that may be produced at the high energies of the early universe, thus solving the monopole problem.
- The shrining of the Hubble radius during inflation allows regions that are currently causally disconnected to have been in causal contact before inflation, thus solving the horizon problem.
- Similarly, quantum fluctuations are strectched to super-horizon scales due to the shrinking Hubble radius, undergoing decoherence and becoming the classical perturbations that seed structure formation in the universe.
- This mechanism not only provides a way to generate primordial density perturbations, but also naturally predicts the nearly scale-invariant power spectrum with Gaussian statistics and the adiabatic feature of initial conditions experimentally observed in the CMB and LSS data.

1.1 The horizon problem

Assuming standard big bang cosmology with a radiation-dominated early universe we find the angle subtended by the particle horizon at the last scattering surface

$$\theta_{\text{hor}} = \frac{d_{\text{hor}}(t_{\text{ls}})}{d_A(t_{\text{ls}})} = \frac{a_{ls}\int_0^{a_l s} c d\tau}{a_{ls}\int_{a_{ls}}^{a_0} c d\tau} = \frac{\int_0^{a_{ls}} \frac{da}{a\mathcal{H}}}{\int_{a_{ls}}^{a_0} \frac{da}{a\mathcal{H}}} \simeq \frac{\frac{a_{eq}}{H_0\sqrt{\Omega_{r0}}} + \frac{2}{H_0\sqrt{\Omega_{m0}}}\left(a_{ls}^{1/2} - a_{eq}^{1/2}\right)}{\frac{2}{H_0\sqrt{\Omega_{m0}}}\left(a_0^{1/2} - a_{ls}^{1/2}\right)} \simeq 2^\circ.$$

This poses a serious problem since the CMB temperature is observed to be isotropic to order 10^{-5} across the entire sky. How could photons be thermalized across $360^\circ \gg 2^\circ$ if they were never in causal contact with each other?

If instead we assume a period of accelerated expansion from $a_i \simeq 0$ to some time a_r before radiation era, the above computation is modified to

$$\theta_{\text{hor}} = \ldots = \frac{\int_0^{a_{ls}} \frac{da}{a\mathcal{H}}}{\int_{a_{ls}}^{a_0} \frac{da}{a\mathcal{H}}} \simeq \frac{\int_{a_i}^{a_r} c d\tau + \frac{a_{eq}-a_r}{H_0\sqrt{\Omega_{r0}}} + \frac{2}{H_0\sqrt{\Omega_{m0}}}\left(a_{ls}^{1/2} - a_{eq}^{1/2}\right)}{\frac{2\left(a_0^{1/2} - a_{ls}^{1/2}\right)}{H_0\sqrt{\Omega_{m0}}}}. \tag{8.1}$$

Recalling $H_{\approx}H_* =$ const and assuming inflation lasts long enough $a_r \gg a_i$, we compute

$$\int_{a_i}^{a_r} \frac{da}{a\mathcal{H}} = \frac{1}{a_r H_*}\left(\frac{a_r}{a_i} - 1\right) \simeq \frac{1}{a_r H_*}\frac{a_r}{a_i} = \frac{1}{a_r H_*} e^{N_e},$$

where N_e is *defined* as the number of e-folds of inflation. The first term in (8.1) dominates, and we simply enforce the causal angle at the last scattering surface to be larger than $2\pi = 360^\circ$ to get a lower bound on the number of inflation e-folds

$$\theta_{\text{hor}} =\simeq \frac{\int_{a_i}^{a_r} c d\tau}{\frac{2\left(a_0^{1/2} - a_{ls}^{1/2}\right)}{H_0\sqrt{\Omega_{m0}}}} \simeq \frac{\frac{1}{a_r H_*} e^{N_e}}{\frac{2\left(a_0^{1/2} - a_{ls}^{1/2}\right)}{H_0\sqrt{\Omega_{m0}}}} \overset{!}{\gtrsim} 2\pi \quad \Rightarrow \quad e^{N_e} \gtrsim 4\pi\,(a_r H_*)\frac{\left(a_0^{1/2} - a_{ls}^{1/2}\right)}{H_0\sqrt{\Omega_{m0}}}. \tag{8.2}$$

To get actual numbers we can assume inflation ends roughly at the GUT scale $H_* \simeq 10^{16}$GeV, and that reheating is efficient enough that radiation dominated era starts immediately afterwards. We then estimate the reheating temperature T_r from the Friedmann equation at the end of inflation

$$H^2 \simeq \frac{8\pi G}{3}\rho \simeq \frac{8\pi G}{3}\frac{\pi^2}{30} g_* T^4 \quad \Rightarrow \quad T_{\text{r}} \simeq \left(\frac{g_* 90}{\pi^3}\right)^{\frac{1}{4}} (H_* M_{\text{Pl}})^{\frac{1}{2}} \simeq 10^{18}\text{GeV}.$$

Since radiation temperature always scales as $T \propto a^{-1}$, we estimate

$$a_r \simeq a_0 \frac{T_0}{T_r} \simeq \frac{2.7K}{10^{18}\text{GeV}} \sim 10^{-30}.$$

Plugging numbers into (8.2), recalling $H_0 \simeq 10^{-42}$GeV, $\Omega_{m0} \simeq 0.3$, $a_{ls} \simeq 10^{-3}$, we find the minimum number of e-folds required

$$e^{N_e} \gtrsim 4\pi\,(a_r H_*)\frac{\left(a_0^{1/2} - a_{ls}^{1/2}\right)}{H_0\sqrt{\Omega_{m0}}} \sim 10^{30} \quad \Rightarrow \quad N_e \gtrsim 60 - 70.$$

1.2 The flatness problem

Consider the density parameters of species relative to the time-dependent critical density $\rho_c(t) = \frac{3H^2(t)}{8\pi G}$,

$$\Omega_r(a) = \frac{\Omega_{r0}}{H^2 a^4}, \quad \Omega_m(a) = \frac{\Omega_{m0}}{H^2 a^3}, \quad \Omega_\Lambda(a) = \frac{\Omega_{\Lambda 0}}{H^2}, \quad \Omega_k(a) = \frac{\Omega_{k0}}{H^2 a^2} = -\frac{k}{H^2 a^2}.$$

The currenlty measured value for $\Omega_{k0} = 0.0007 \pm 0.0019$ is consistent with a spatially flat universe. When a species x dominates we have $H^2 \propto a^{-3(1+w_x)}$, and curvature density parameter goes like $\Omega_k(a) \propto a^{1+3w_x}$ over different epochs. Ignoring vacuum energy only relevant at late times, we find that $\Omega_k(a)$ has always been growing over time up to the tiny value Ω_{k0} observed today. Either the universe is exactly flat $\Omega_{k0} \equiv 0$, which seems to require some explanation, or it must have started with an extremely small value of $\Omega_k(a) \sim a^{1+3w_x}\Omega_{k0}$ in the early universe so as to match the present value, which similarly requires estreme fine tuning.

If instead a species with $w_x < -1/3$ dominated the early universe, we have accelerated expansion, the Hubble radius shrinks and the curvature density parameter $\Omega_k(a)$ decreases over time, solving the flateness problem. In the case of approximately de Sitter expansion $H \simeq$ const, i.e. $w_x \sim -1$, we find

$$\Omega_k(a) = \frac{\Omega_{k0}}{H^2 a^2} \simeq \frac{\Omega_{k0}}{H^2} e^{-2Ht} \to 0 \quad \text{as} \quad t \to \infty.$$

1.3 The exotic monopole problem

Grand Unified Theories generically predict the production of topologically stable magnetic monopoles during symmetry-breaking phase transitions in the early universe via the Kibble mechanism. Since these objects are massive and behave as non-relativistic matter, their energy density scales as $\rho \propto a^{-3}$ and their energy density would rapidly dominate the universe, leading to a relic abundance vastly exceeding observational bounds. In standard Big Bang cosmology this constitutes the monopole problem. If instead a period of accelerated expansion occurs before or during monopole production, the exponential growth of the scale factor dilutes their number density as $n_M \propto e^{-3N_e}$, rendering their present abundance negligibly small and thereby solving the monopole problem.

1.4 Seeding primordial perturbations: scale invariance, Gaussianity, and adiabaticity

Adiabatic initial conditions with single clock inflation

This assumption, motivated by inflation, ensures that any departure between species arises only from their subsequent dynamics rather than from different initial phases. Concretely, adiabatic perturbations enforce that the local state of matter (determined for example by energy density and pressure) at some spacetime point $(\tau, \bar{x})$ of the perturbed universe is the same as in the background universe at some slightly different time $\tau + \delta\tau(\bar{x})$, that is

$$\delta\rho_\ell(\tau, \boldsymbol{x}) \equiv \bar{\rho}_\ell(\tau + \delta\tau(\boldsymbol{x})) - \bar{\rho}_\ell(\tau) = \dot{\bar{\rho}}_\ell\, \delta\tau(\boldsymbol{x}),$$

where the local time shift $\delta\tau(x)$ is the same for all species, and analogously for δP and similar quantities. Then, introducing the fractional density contrast $\delta_\ell \equiv \frac{\delta\rho_\ell}{\bar{\rho}_\ell}$, this implies that

$$\delta\tau(x) = \frac{\bar{\rho}_\ell}{\dot{\bar{\rho}}_\ell}\,\delta_\ell = \frac{\bar{\rho}_{\ell'}}{\dot{\bar{\rho}}_{\ell'}}\,\delta_{\ell'} \qquad \text{for all species } \ell \text{ and } \ell'.$$

In particular, for species x, x' whose energy density separetely conserved $\nabla_\mu T^\mu_{(x)\,0} = \nabla_\mu T^\mu_{(x')\,0} = 0$ we get the relation

$$\frac{\delta_x}{1+w_x} = \frac{\delta_{x'}}{1+w_{x'}} = \frac{\bar{\rho}_\ell}{\dot{\bar{\rho}}_\ell}\,\delta_\ell \qquad \text{for all other species } \ell. \tag{8.3}$$

Crucially, for adiabatic fluctuations the total density perturbation is dominated at any time by the species dominating the background

$$\delta\rho_{\rm tot} := \bar{\rho}_{\rm tot}\delta_{\rm tot} = \sum_\ell \bar{\rho}_\ell\delta_\ell \sim \bar{\rho}_{\rm dom}\delta_{\rm dom} \qquad \text{for adiabatic fluctuations,}$$

since by (8.3) all the δ_ℓ are comparable.

2 Decoherence of quantum fluctuations during inflation

finish

We now explicitly derive the EoM for the inflaton and its perturbations, and show how the quantum fluctuations of the inflaton field are stretched to super-horizon scales during inflation, undergoing decoherence and becoming the classical perturbations that seed structure formation in the universe.

3 Initial conditions for the Einstein-Boltzmann system

finish

We now explicitly derive the initial conditions for the Einstein-Boltzmann system of equations from the conservation of the comoving curvature perturbation $\mathcal{R}$ on super-horizon scales and a suitable analysis of the equations in the superhorizon limit

finish cf Whinter 16 & baumann book

9 Structure Formation

Metric perturbations remain small
Treating metric perturbations to linear order is justified since they remain very small, even when energy-density perturbations become extremely non-linear. For example, compare the (dimensionless) gravitational potential[1] $\phi \sim \frac{GM}{R}$ and the density contrast $\delta\rho/\bar{\rho}$, relative to the cosmic average matter density today, for various astrophysical objects:

- Earth: $\phi \sim 10^{-9}, \quad \frac{\delta\rho}{\rho} \sim 10^{30}$
- Sun: $\phi \sim 10^{-6}, \quad \frac{\delta\rho}{\rho} \sim 10^{28}$
- Milky Way: $\phi \sim 10^{-6}, \quad \frac{\delta\rho}{\rho} \sim 10^{6}$
- Galaxy cluster: $\phi \sim 10^{-5}, \quad \frac{\delta\rho}{\rho} \sim 10^{3}$
- Black hole at Schwarzschild radius $R_S = 2GM$: $\phi \sim \frac{1}{2}, \quad \frac{\delta\rho}{\rho} \sim \infty$.

1 Newtonian theory

2 Structure formation across epochs

3 Correlation functions, power spectrum, and BAO

The fit of the matter correlation function $\xi(r)$ to the observed galaxy correlation function $\xi^{\text{obs}}(r)$ is one of the most important cosmological observables, and it is used to constrain cosmological parameters. The best fit is

$$\xi(r) \simeq \left(\frac{r}{r_0}\right)^{-\gamma} + \text{BAO feature at } r \sim 150\text{Mpc},$$

where $r_0 \sim 5.4$Mpc and $\gamma \sim 1.8$. Suitable cutoffs are inserted to remove the unphysical divergence at $r = 0$ and to account for the finite size of the universe at large r.

[1] Identified with the Newtonian-gauge potential Φ

10 Full Shape & Nonlinear Theory

1 Redshift space

Consider two points in the universe separated by a cmoving distance χ. The source emits an electromagnetic signal with frequency ω_e and wavelength λ_e. The signal is observed with frequency ω_o and wavelength λ_o. If the two observers are comoving i.e. fixed comoving coordinates, the comoving distance travelled by the photon stays constant. Since the photon travels along a null geodesic $ad\chi = dt$, therefore we find

$$\chi = \int_{\chi_e}^{\chi_o} d\chi = \int_{t_e}^{t_o} \frac{dt}{a(t)} = \int_{t_e+\Delta t_e}^{t_o+\Delta t_o} \frac{dt}{a(t)} \quad \Rightarrow \quad \int_{t_e}^{t_e+\Delta t_e} \frac{dt}{a(t)} = \int_{t_o}^{t_o+\Delta t_o} \frac{dt}{a(t)}.$$

If the time intervals Δt_e and Δt_o are small, we can approximate $a(t)$ as constant over these intervals, and we find

$$\frac{\Delta t_o}{a(t_o)} = \frac{\Delta t_e}{a(t_e)} \quad \Rightarrow \quad \frac{\omega_o}{\omega_e} = \frac{\lambda_e}{\lambda_o} = \frac{a(t_e)}{a(t_o)} =: 1+z.$$

More generally, for $1+z := \frac{a_*}{a}$ with $a_* = 1$ today, we have

$$d\chi = \frac{dt}{a} = \frac{da}{a^2 H(a)} = -\frac{dz}{H(z)}, \quad \chi = \int_{t_e}^{t_o} \frac{dt}{a(t)} = \int_{a_e}^{a_o} \frac{da}{a^2 H(a)} = \int_{z_o}^{z_e} \frac{dz}{H(z)} = \int_{z_o}^{z_e} \frac{dz}{H(z)}.$$

If $H(z)$ varies slowly with z between z_o and z_e, we find back Hubble's law $\Delta z \simeq H(z_o)\chi$.

Now consider a source moving with peculiar velocity $\vec{v}$. The observed redshift is then given by a combination of the cosmological redshift and the Doppler shift, which are multiplicative simply because $\lambda_2/\lambda_0 = (\lambda_2/\lambda_1)(\lambda_1/\lambda_0)$,

$$1+z = (1+z_\mathrm{c})(1+z_\mathrm{d}) = 1 + z_\mathrm{c} + (1+z_\mathrm{c})z_\mathrm{d}.$$

For small velocities $\beta \ll 1$, regardless of the relative angle between the velocity vector and the line of sight $\hat{n}$, the relativistic Doppler shift is approximated to first order as

$$\frac{\lambda_o}{\lambda_e} = \frac{\omega_e}{\omega_o} = \frac{1}{\gamma\,(1-\beta\hat{v}\cdot\hat{n})} \approx 1 + \frac{\vec{v}\cdot\hat{n}}{c} + O(\beta^2).$$

The actual comoving distance χ in real space and the inferred comoving distance χ_s in redshift space are then related by

$$\chi_s = \int_0^z \frac{dz}{H(z)} = \int_0^{z_\mathrm{c}} \frac{dz}{H(z)} + \int_{z_\mathrm{c}}^{z_\mathrm{c}+(1+z_\mathrm{c})z_d} \frac{dz}{H(z)} \approx \chi + \frac{(1+z_\mathrm{c})z_\mathrm{d}}{H(z_\mathrm{c})} \approx \chi + \frac{(1+z_\mathrm{c})}{H(z_\mathrm{c})}\frac{\vec{v}\cdot\hat{n}}{c}.$$

For present times comoving and physical distance coincide since $a_0 = 1$, so that we can replace χ with r and write

$$\vec{r}_s = \vec{r} + \frac{(1+z)}{H(z)}\frac{\vec{v}\cdot\hat{n}}{c}\hat{n}.$$

Let us now relate the density contrast in redshift space δ_s to the density contrast in real space δ. Since total number of galaxies is conserved and the backround density $\bar{\rho}$ is homogeneous, we have

$$\bar{n}(1+\delta_s(\vec{r}_s))d^3\vec{r}_s = \bar{n}(1+\delta(\vec{r}))d^3\vec{r} \quad \Rightarrow \quad 1+\delta_s(\vec{r}_s) = (1+\delta(\vec{r}))\left|\frac{\partial\vec{r}}{\partial\vec{r}_s}\right|.$$

We now compute... Finish

2 Spherical Collapse

3 Press-Schechter Formalism and Mass Functions

4 Halo Model

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Galactic & Extragalactic Astronomy

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Part VII

Appendices

A Matheatical tools

1 Gaussian random fields

Check & finish

A Gaussian random field is a stochastic process

$$X : \Omega \times U \to F \mid \omega, u \mapsto X(\omega, u),$$

such that for any finite collection of parameters $u_1, \ldots, u_n \in U$ the random vector $(X(\cdot, u_1), \ldots, X(\cdot, u_n))$ is Gaussian distributed in F^n. Crucially, the elements of a gaussian random vectors are independent if and only if they are uncorrelated, i.e. if the off-diagonal elements of the covariance matrix vanish.

Quantum mechanical processes naturally induce nearly Gaussian random fields, since free quantum fields are a collection of independent harmonic oscillators and interactions are small perturbations thereof. The wave function of each oscillator is a Gaussian times Hermite polynomials, or simply a Gaussian for the ground state, so that the field wave function is simply the product of these wave functions, which is again a Gaussian up to polynomial factors.

Another key feature is that a Gaussian random field is always ergodic, since the covariance function decays to zero at large distances, so that the law of large numbers applies and ensemble averages can be replaced with parameter averages.

A key feature of Gaussian random fields is that they are completely determined by their mean and covariance functions. In particular, all reduced n-point functions vanish and all n-point functions can be expressed in terms of the 2-point function via Wick's theorem.

$$\langle X(\cdot, u_1) \cdots X(\cdot, u_n) \rangle_\Omega = \sum_{\text{pairings}} \prod_{\text{pairs } (i,j)} \langle X(\cdot, u_i) X(\cdot, u_j) \rangle_\Omega.$$

In particular homogeneity requires the mean function is constant $\langle X(\cdot, u) \rangle_\Omega = \mu$, and isotropy further requires the covariance function depends only on the distance $|u - u'|$ between parameters $\langle X(\cdot, u) X(\cdot, u') \rangle_\Omega = \xi(|u - u'|)$. For simplicity we subtract off the constant mean μ and consider real field $\delta \in \mathbb{R}$ in the discussion below.

A Gaussian random field in real space is a Gaussian random field in Fourier space and viceversa. Notably, if the correlation function in real space is homogeneous $\xi(x, y) = \xi(x - y)$, then the Fourier modes are independent, i.e. the covariance function in Fourier space is always diagonal

$$\langle \delta_{\vec{k}} \delta_{\vec{k}'} \rangle_\Omega = (2\pi)^3 \delta^{(3)}(\vec{k} + \vec{k}') P(\vec{k}), \quad \text{where} \quad P(\vec{k}) := \int d^3\vec{x} e^{-i\vec{k}\cdot\vec{x}} \xi(\vec{x}).$$

This is simply expressing the general fact that spatial homogeneity implies total momentum conservation. The probability distibution of a single mode is a *complex* Gaussian, since linear combination of Gaussians remain Gaussian, with variance given by the power spectrum $P(\vec{k})$ and *uniformly distributed* random phase

$$\begin{aligned}
\mathbb{P}(\delta_{\vec{k}})\, d^2_{\mathbb{C}}\delta_{\vec{k}} &= \frac{1}{\sqrt{2\pi P(\vec{k})}} \exp\left(-\frac{|\delta_{\vec{k}}|^2}{2P(\vec{k})}\right) d^2_{\mathbb{C}}\delta_{\vec{k}} \\
&= \frac{|\delta_{\vec{k}}|}{\sqrt{2\pi P(\vec{k})}} \exp\left(-\frac{|\delta_{\vec{k}}|^2}{2P(\vec{k})}\right) d|\delta_{\vec{k}}|\, d\arg(\delta_{\vec{k}}).
\end{aligned}$$

Since the field $\delta(\vec{x})$ is real then $\delta^*_{\vec{k}} = \delta_{-\vec{k}}$. The probability distribution of a given set of Fourier modes is just the product of the single mode distributions

$$\begin{aligned}
\mathbb{P}(\delta_{\vec{k}_1} \ldots \vec{k}_n) \prod_{j=1}^n d^2_{\mathbb{C}}\delta_{\vec{k}_j} &= \prod_{j=1}^n \mathbb{P}(\delta_{\vec{k}_j})\, d^2_{\mathbb{C}}\delta_{\vec{k}_j} \\
&= \prod_{i=1}^n \frac{1}{\sqrt{2\pi P(k_i)}} \exp\left(-\frac{|\delta_{\vec{k}_i}|^2}{2P(k_i)}\right) d^2_{\mathbb{C}}\delta_{\vec{k}_i} = \prod_{i=1}^n \frac{1}{\sqrt{2\pi P(k_i)}} \exp\left(-\frac{|\delta_{\vec{k}_i}|^2}{2P(k_i)}\right) k_i^2 dk_i d\Omega_i.
\end{aligned}$$

If isotropy also holds i.e. $\xi(x,y) = \xi(|x-y|)$, the joint law of Fourier modes $\delta_{\vec{k}}$ only depend on the magnitude of momenta $k = |\vec{k}|$ and $P(\vec{k}) \equiv P(k)$ is independent of the direction $\hat{k}$. This is the case in Cosmology.

2 Spherical harmonics & Legendre polynomials

Finish

Angular scale of variation of spherical harmonics

Tha spherical harmonic $Y_{\ell m}(\theta,\phi)$ has an angular scale of variation $\Delta\theta \sim \pi/\ell$ in the θ direction and $\Delta\phi \sim 2\pi/m$ in the ϕ direction. The latter is immediately understood since the ϕ dependence is of the form $Y_{\ell m}(\theta,\phi) = F_{\ell m}(\theta)e^{im\phi}$, so that the function is periodic in ϕ with period $2\pi/m$. The former is less trivial, it can be understood by looking at the explicit form of $Y_{\ell m}(\theta,\phi)$ in terms of associated Legendre polynomials $P_\ell^m(\cos\theta)$, which are polynomials of degree ℓ in $\cos\theta$ and thus have ℓ roots in the interval $[-1,1]$, corresponding to ℓ zeros of $Y_{\ell m}(\theta,\phi)$ in the interval $\theta \in [0,\pi]$, so that the typical distance between zeros is π/ℓ. Naively we would thus say the wavelength of such a mode is $\lambda \sim \frac{2\pi}{\ell}$, so that the typical angular size of a CMB hot or cold spot (i.e. of a peak or trough of the wave) is roughly π/ℓ.

Alternatively focusing on the higher harmonics $Y_{\ell\ell} = (\sin\theta)^\ell e^{i\ell\phi}$, so that

$$\frac{1}{Y_{\ell\ell}}\frac{\partial Y_{\ell\ell}}{\partial\theta}\Delta\theta = \ell\cot\theta\Delta\theta,$$

so that the angular scale of variation in the θ direction is $\Delta\theta \sim 1/(\ell\cot\theta)$, which is of order π/ℓ for θ not too close to 0 or π.

Spherical harmonics

Start from the decomposition

$$L = x \wedge p = -i\hbar(x\wedge\nabla)$$

Then

$$-\frac{L^2}{\hbar^2} = (x\wedge\nabla)\cdot(x\wedge\nabla) = x_a\partial_b x_a\partial_b - x_a\partial_b x_b\partial_a = r^2\partial^2 - x_a x_b\partial_a\partial_b - 2x_a\partial_a$$
$$\Rightarrow \quad \nabla^2 = -\frac{L^2}{\hbar^2 r^2} + \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right).$$

The angular momentum components are

$$L_x = -i\hbar(y\partial_z - z\partial_y) = L_y \qquad = -i\hbar(z\partial_x - x\partial_z)L_z = -i\hbar(x\partial_y - y\partial_x) = -i\hbar\frac{\partial}{\partial\phi}.$$

The raising and lowering operators are defined as

$$L_\pm = L_x \pm iL_y = \pm e^{\pm i\phi}\left(\frac{\partial}{\partial\theta} \pm i\cot\theta\frac{\partial}{\partial\phi}\right), \quad \text{i.e.} \quad L_- = L_+^\dagger.$$

We originally introduced angular momentum via the orbital angular momentum operators $\mathbf{J} = \mathbf{L} = \mathbf{X}\wedge\mathbf{P}$ in (**??**). On the other hand, we know that $\mathbf{X}$ and $\mathbf{P}$ act on wave functions $\psi(\mathbf{x})$ as multiplication by $\mathbf{x}$ and $-\mathrm{i}\hbar\nabla$, respectively, which makes the $\mathbf{L}$ differential operators acting on such wave functions. For example,

$$L_3 = X_1P_2 - X_2P_1 = -\mathrm{i}\hbar\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right)$$

writing $\mathbf{x} = (x,y,z)$. As for the harmonic oscillator it must then be possible to find explicit eigenfunctions for these operators, representing the eigenstates of angular momentum we found abstractly in the previous subsection. It is convenient to first write the operators L_i in terms of spherical polar coordinates $(x,y,z) = (r\sin\theta\cos\phi, r\sin\theta\sin\phi, r\cos\theta)$, and then using the chain rule

$$\frac{\partial}{\partial\phi} = \frac{\partial x}{\partial\phi}\frac{\partial}{\partial x} + \frac{\partial y}{\partial\phi}\frac{\partial}{\partial y} + \frac{\partial z}{\partial\phi}\frac{\partial}{\partial z} = -y\frac{\partial}{\partial x} + x\frac{\partial}{\partial y}$$

Thus

$$L_3 = -\mathrm{i}\hbar\frac{\partial}{\partial\phi}$$

Recall that the eigenstates ψ_m defined in the previous subsection had eigenvalue $m\hbar$ under $J_3 = L_3$, so that $L_3\psi_m = m\hbar\psi_m$ reads

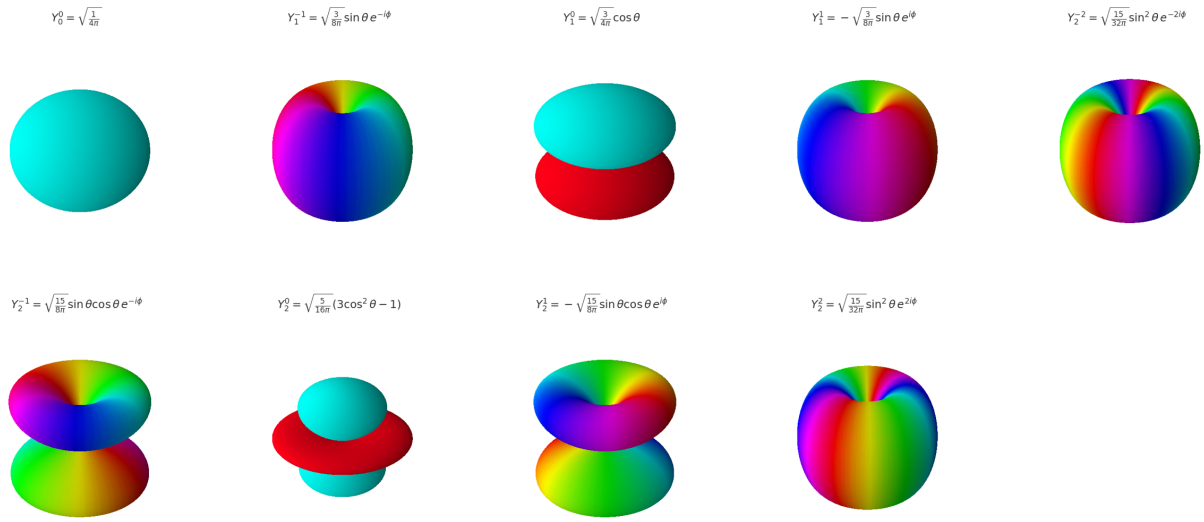
$$\frac{\partial\psi_m}{\partial\phi} = \mathrm{i}m\psi_m$$

which integrates to

$$\psi_m(r,\theta,\phi) = F(r,\theta)\mathrm{e}^{\mathrm{i}m\phi}$$

The coordinate ϕ has period 2π, so in order for the function (**??**) to be single-valued (well defined) we must have $m \in \mathbb{Z}$ being an integer. Correspondingly then also $j \in \mathbb{Z}_{\geqslant 0}$ must be integer in Theorem 8.6. This proves **Proposition 8.8** For orbital angular momentum the parameters j, m in Theorem 8.6 must both be integers. In this context we instead label $j = \ell \in \mathbb{Z}_{\geqslant 0}$. To pursue this further, let us also work out the raising and lowering operators in terms of spherical polars. **Proposition 8.9** In spherical polar coordinates the raising and lowering operators are

$$L_+ = L_1 + \mathrm{i}L_2 = \hbar\mathrm{e}^{\mathrm{i}\phi}\left(\frac{\partial}{\partial\theta} + \mathrm{i}\cot\theta\frac{\partial}{\partial\phi}\right)$$
$$L_- = L_1 - \mathrm{i}L_2 = -\hbar\mathrm{e}^{-\mathrm{i}\phi}\left(\frac{\partial}{\partial\theta} - \mathrm{i}\cot\theta\frac{\partial}{\partial\phi}\right)$$



2.1 Legendre polynomials

Legendre polynomials are simply defined as spherical harmonics with $m = 0$ up to a normalization factor:

$$P_\ell(\cos\theta) = \sqrt{\frac{4\pi}{2\ell+1}}Y_{\ell 0}(\theta,\phi).$$

Since spherical harmonics are eigenfunctions of the angular momentum operators, Legendre polynomials are also eigenfunctions of the L^2 operator. Since $\{Y_{\ell m}\}$ form a CONS of $L^2(S^2)$, the set $\{P_\ell\}$ also form a CONS of the subspace of $L^2(S^2)$ consisting of functions independent of ϕ or equivalently of $L^2([-1,1], d\cos\theta)$. Since $Y^*_{\ell,m} = (-1)^m Y_{\ell,-m}$, Legendre polynomials are real $P^*_\ell = P_\ell$. From the orthonormality of spherical harmonics $\int_{S^2} Y^*_{\ell m} Y_{\ell' m'} d\phi d\cos\theta = \delta_{\ell\ell'}\delta_{mm'}$ we derive the orthogonality relation for Legendre polynomials:

$$\int_{-1}^{1} P_\ell(\mu)P_{\ell'}(\mu)d\mu = \frac{2}{2\ell+1}\delta_{\ell\ell'}.$$

We then expand functions $f(\mu)$ in Legendre polynomials, where it is ofetn convenient to absorb powers of i in the expansion coefficients,

$$f(\mu) = \sum_{\ell=0}^{\infty} \frac{2\ell+1}{i^\ell} f_\ell P_\ell(\mu), \quad \text{where} \quad f_\ell = \frac{i^\ell}{2} \int_{-1}^{1} f(\mu) P_\ell(\mu) d\mu.$$

They satisfy the following recursion relations:

$$\mu P_\ell(\mu) = \frac{\ell+1}{2\ell+1} P_{\ell+1}(\mu) + \frac{\ell}{2\ell+1} P_{\ell-1}(\mu), \quad \text{and} \quad \frac{dP_\ell}{d\mu} = \frac{\ell}{1-\mu^2}(P_{\ell-1}(\mu) - \mu P_\ell(\mu)).$$

The first 3 polynomials are

$$P_0(\mu) = 1, \quad P_1(\mu) = \mu, \quad P_2(\mu) = \frac{1}{2}(3\mu^2 - 1).$$

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